

Bd. 13 - Ganze Zahlen

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Kapitel 1

Einleitung

Die ganzen Zahlen entstehen aus Paaren natürlicher Zahlen. Das Paar (a, b) wird als formale Differenz $a - b$ gelesen. Da verschiedene Paare dieselbe Differenz beschreiben können, werden genau diejenigen Paare identifiziert, deren Kreuzsummen übereinstimmen. Die allgemeine Theorie der Äquivalenzrelationen und Quotienten aus Band 10 liefert anschließend die Menge der ganzen Zahlen.

Nach der Konstruktion werden Nachfolger, Vorgänger und die zweiseitige Induktion zunächst im Quotientenmodell bewiesen. Erst danach werden diese Eigenschaften als erweiterte Peano-Axiome registriert. Addition, Multiplikation und Negation werden stets erst nach einem Beweis ihrer Repräsentantenunabhängigkeit eingeführt.

Kapitel 2

Konstruktion der ganzen Zahlen

2.1 Differenzenpaare

Definition 13.2.1.1 (Träger der Differenzenpaare).

Mengen: $D_{\mathbb{Z}}$
Neue Symbole: $D_{\mathbb{Z}}$

$$D_{\mathbb{Z}} := \mathbb{N} \times \mathbb{N}$$

Definition 13.2.1.2 (Äquivalenz von Differenzenpaaren).

Mengen: a, b, c, d
Relationsoperatoren: $\sim_{\mathbb{Z}}$
Trägermenge: $D_{\mathbb{Z}}$
Neue Symbole: $\sim_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash (a, b) \sim_{\mathbb{Z}} (c, d) := a + d = b + c$$

Theorem 13.2.1.1 (Charakterisierung des Äquivalenzoperators auf Differenzenpaaren).

Mengen: a, b, c, d
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash (a, b) \sim_{\mathbb{Z}} (c, d) \leftrightarrow a + d = b + c$$

$$\begin{array}{l} 1 \quad 1 \quad a, b, c, d \in \mathbb{N} \quad A \\ 1 \quad 2 \quad (a, b) \sim_{\mathbb{Z}} (c, d) \leftrightarrow a + d = b + c \quad \text{Def. 13.2.1.2(1)} \end{array}$$

Theorem 13.2.1.2 (Reflexivität auf Differenzenpaaren).

Mengen: a, b
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$a, b \in \mathbb{N} \vdash (a, b) \sim_{\mathbb{Z}} (a, b)$$

$$\begin{array}{l} 1 \ 1 \ a, b \in \mathbb{N} \quad A \\ 1 \ 2 \ a + b = b + a \quad \text{Th. 12.4.4.8(1)} \\ 1 \ 3 \ (a, b) \sim_{\mathbb{Z}} (a, b) \quad \text{Th. 13.2.1.1(1,2)} \end{array}$$

Theorem 13.2.1.3 (Symmetrie auf Differenzenpaaren).

Mengen: a, b, c, d
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N}, (a, b) \sim_{\mathbb{Z}} (c, d) \vdash (c, d) \sim_{\mathbb{Z}} (a, b)$$

$$\begin{array}{l} 1 \ 1 \ a, b, c, d \in \mathbb{N} \quad A \\ 2 \ 2 \ (a, b) \sim_{\mathbb{Z}} (c, d) \quad A \\ 1,2 \ 3 \ a + d = b + c \quad \text{Th. 13.2.1.1(1,2)} \\ 1,2 \ 4 \ c + b = b + c \quad \text{Th. 12.4.4.8(1)} \\ 1,2 \ 5 \ b + c = a + d \quad \text{Th. 2.9.1.1(3)} \\ 1,2 \ 6 \ a + d = d + a \quad \text{Th. 12.4.4.8(1)} \\ 1,2 \ 7 \ c + b = d + a = E(4, 5, 6) \\ 1,2 \ 8 \ (c, d) \sim_{\mathbb{Z}} (a, b) \quad \text{Th. 13.2.1.1(1,7)} \end{array}$$

Theorem 13.2.1.4 (Transitivität auf Differenzenpaaren).

Mengen: a, b, c, d, e, f
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$\begin{array}{l} a, b, c, d, e, f \in \mathbb{N}, (a, b) \sim_{\mathbb{Z}} (c, d), \\ (c, d) \sim_{\mathbb{Z}} (e, f) \vdash (a, b) \sim_{\mathbb{Z}} (e, f) \end{array}$$

Beweis.

Zusammenführen der Kreuzsummen

Th. 13.2.1.4(i)

$$\begin{array}{l} a, b, c, d, e, f \in \mathbb{N}, a + d = b + c, c + f = d + e \vdash \\ (a + f) + (c + d) = (b + e) + (c + d) \end{array}$$

1	$a, b, c, d, e, f \in \mathbb{N}$	A
2	$a + d = b + c$	A
3	$c + f = d + e$	A
1,2,3	$(a + d) + (c + f) = (b + c) + (d + e) = E(2, 3)$	
1	$(a + d) + (c + f) = (a + f) + (c + d)$	Th. 12.4.4.10(1)
1	$(b + c) + (d + e) = (b + e) + (c + d)$	Th. 12.4.4.10(1)
1,2,3	$(a + f) + (c + d) = (b + e) + (c + d) = E(4, 5, 6)$	

Kürzen der gemeinsamen Summe

Th. 13.2.1.4(ii)

$$a, b, c, d, e, f \in \mathbb{N}, (a + f) + (c + d) = (b + e) + (c + d)$$

$$\vdash a + f = b + e$$

1	$a, b, c, d, e, f \in \mathbb{N}$	A
2	$(a + f) + (c + d) = (b + e) + (c + d)$	A
1	$a + f \in \mathbb{N}$	Th. 12.4.4.1(1)
1	$b + e \in \mathbb{N}$	Th. 12.4.4.1(1)
1	$c + d \in \mathbb{N}$	Th. 12.4.4.1(1)
1,2	$a + f = b + e$	Th. 12.4.5.30(3,4,5,2)

Schluss:

1	$a, b, c, d, e, f \in \mathbb{N}$	A
2	$(a, b) \sim_{\mathbb{Z}} (c, d)$	A
3	$(c, d) \sim_{\mathbb{Z}} (e, f)$	A
1,2	$a + d = b + c$	Th. 13.2.1.1(1,2)
1,3	$c + f = d + e$	Th. 13.2.1.1(1,3)
1,2,3	$(a + f) + (c + d) = (b + e) + (c + d)$	Th. 13.2.1.4(i)(1,4,5)
1,2,3	$a + f = b + e$	Th. 13.2.1.4(ii)(1,6)
1,2,3	$(a, b) \sim_{\mathbb{Z}} (e, f)$	Th. 13.2.1.1(1,7)

□

Theorem 13.2.1.5 (Der Äquivalenzoperator auf Differenzenpaaren ist eine Äquivalenzrelation).

Mengen: $D_{\mathbb{Z}}$
Relationsoperatoren: $\sim_{\mathbb{Z}}$

$\text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}})$

Beweis.

Reflexivität

Th. 13.2.1.5(i)

$$x \in D_{\mathbb{Z}} \vdash x \sim_{\mathbb{Z}} x$$

$$\begin{array}{ll}
1 \quad 1 \quad x \in D_{\mathbb{Z}} & A \\
1 \quad 2 \quad \exists a \in \mathbb{N} \exists b \in \mathbb{N} x = (a, b) & \text{Th. 3.16.5.4(1)} \\
3 \quad 3 \quad a \in \mathbb{N} \wedge b \in \mathbb{N} \wedge x = (a, b) & A \\
3 \quad 4 \quad (a, b) \sim_{\mathbb{Z}} (a, b) & \text{Th. 13.2.1.2(3)} \\
3 \quad 5 \quad x \sim_{\mathbb{Z}} x & = E(3, 4) \\
1 \quad 6 \quad x \sim_{\mathbb{Z}} x & \exists E(2, 3, 5)
\end{array}$$

Symmetrie

Th. 13.2.1.5(ii)

$$x \sim_{\mathbb{Z}} y \vdash y \sim_{\mathbb{Z}} x$$

$$\begin{array}{ll}
1 \quad 1 \quad x \sim_{\mathbb{Z}} y & A \\
1 \quad 2 \quad y \sim_{\mathbb{Z}} x & \text{Th. 13.2.1.3(1)}
\end{array}$$

Transitivität

Th. 13.2.1.5(iii)

$$x \sim_{\mathbb{Z}} y, y \sim_{\mathbb{Z}} z \vdash x \sim_{\mathbb{Z}} z$$

$$\begin{array}{ll}
1 \quad 1 \quad x \sim_{\mathbb{Z}} y & A \\
2 \quad 2 \quad y \sim_{\mathbb{Z}} z & A \\
1,2 \quad 3 \quad x \sim_{\mathbb{Z}} z & \text{Th. 13.2.1.4(1,2)}
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 \quad x \in D_{\mathbb{Z}} \rightarrow x \sim_{\mathbb{Z}} x & \text{Th. 13.2.1.5(i)} \\
2 \quad x \sim_{\mathbb{Z}} y \rightarrow y \sim_{\mathbb{Z}} x & \text{Th. 13.2.1.5(ii)} \\
3 \quad 3 \quad x \sim_{\mathbb{Z}} y & A \\
4 \quad 4 \quad y \sim_{\mathbb{Z}} z & A \\
3,4 \quad 5 \quad x \sim_{\mathbb{Z}} z & \text{Th. 13.2.1.5(iii)(3,4)} \\
6 \quad \text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}}) & \text{Def. 10.2.2.1(1,2,5)}
\end{array}$$

□

2.2 Quotient und Klassen

Definition 13.2.2.1 (Menge der ganzen Zahlen).

Mengen: $D_{\mathbb{Z}}, \mathbb{Z}$
Relationsoperatoren: $\sim_{\mathbb{Z}}$
Neue Symbole: $\mathbb{Z}$

$$\mathbb{Z} := D_{\mathbb{Z}} / \sim_{\mathbb{Z}}$$

Definition 13.2.2.2 (Klasse eines Differenzenpaares).

Mengen: $a, b, D_{\mathbb{Z}}$

Relationsoperatoren: $\sim_{\mathbb{Z}}$

Neue Symbole: $[a, b]_{\mathbb{Z}}$

$$a, b \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} := [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$$

Theorem 13.2.2.1 (Klassen sind ganze Zahlen).

Mengen: a, b

$$a, b \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} \in \mathbb{Z}$$

1	1	$a, b \in \mathbb{N}$	A
1	2	$(a, b) \in D_{\mathbb{Z}}$	Th. 3.16.5.5(1)
	3	$\text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}})$	Th. 13.2.1.5
1	4	$[(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} \in D_{\mathbb{Z}} / \sim_{\mathbb{Z}}$	Th. 10.4.2.2(3, 2)
1	5	$[a, b]_{\mathbb{Z}} = [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$	Def. 13.2.2.2(1)
	6	$\mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}}$	Def. 13.2.2.1
1	7	$[a, b]_{\mathbb{Z}} \in \mathbb{Z}$	$= E(5, 6, 4)$

Theorem 13.2.2.2 (Gleichheit von Ganzzahlklassen).

Mengen: a, b, c, d

Relationsoperatoren: $\sim_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow a + d = b + c$$

Beweis.

Klassenkriterium

Th. 13.2.2.2(i)

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow (a, b) \sim_{\mathbb{Z}} (c, d)$$

1	1	$a, b, c, d \in \mathbb{N}$	A
	2	$\text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}})$	Th. 13.2.1.5
1	3	$(a, b) \in D_{\mathbb{Z}}$	Th. 3.16.5.5(1)
1	4	$(c, d) \in D_{\mathbb{Z}}$	Th. 3.16.5.5(1)
1	5	$[(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} = [(c, d)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} \leftrightarrow (a, b) \sim_{\mathbb{Z}} (c, d)$	Th. 10.4.1.7(2, 3, 4)
1	6	$[a, b]_{\mathbb{Z}} = [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$	Def. 13.2.2.2(1)
1	7	$[c, d]_{\mathbb{Z}} = [(c, d)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$	Def. 13.2.2.2(1)

$$1 \quad 8 \quad [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow (a, b) \sim_{\mathbb{Z}} (c, d) \quad = E(5, 6, 7)$$

Schluss:

$$\begin{array}{ll} 1 \quad 1 \quad a, b, c, d \in \mathbb{N} & A \\ 1 \quad 2 \quad [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow (a, b) \sim_{\mathbb{Z}} (c, d) & \text{Th. 13.2.2.2(i)(1)} \\ 1 \quad 3 \quad (a, b) \sim_{\mathbb{Z}} (c, d) \leftrightarrow a + d = b + c & \text{Th. 13.2.1.1(1)} \\ 1 \quad 4 \quad [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \leftrightarrow a + d = b + c & \text{Th. 2.2.3.5(2,3)} \end{array}$$

□

Theorem 13.2.2.3 (Jede ganze Zahl besitzt ein Differenzenpaar).

Mengen: $z, a, b, D_{\mathbb{Z}}$

$$z \in \mathbb{Z} \vdash \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$$

Beweis.

Quotientenrepräsentant

Th. 13.2.2.3(i)

$$z \in \mathbb{Z} \vdash \exists x \in D_{\mathbb{Z}} z = [x]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}}$$

$$\begin{array}{ll} 1 \quad 1 \quad z \in \mathbb{Z} & A \\ 2 \quad \mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}} & \text{Def. 13.2.2.1} \\ 1 \quad 3 \quad z \in D_{\mathbb{Z}} / \sim_{\mathbb{Z}} & = E(2, 1) \\ 4 \quad \text{EqRel}(D_{\mathbb{Z}}, \sim_{\mathbb{Z}}) & \text{Th. 13.2.1.5} \\ 1 \quad 5 \quad \exists x \in D_{\mathbb{Z}} z = [x]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & \text{Th. 10.4.2.4(4,3)} \end{array}$$

Schluss:

$$\begin{array}{ll} 1 \quad 1 \quad z \in \mathbb{Z} & A \\ 1 \quad 2 \quad \exists x \in D_{\mathbb{Z}} z = [x]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & \text{Th. 13.2.2.3(i)(1)} \\ 3 \quad 3 \quad x \in D_{\mathbb{Z}} \wedge z = [x]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & A \\ 3 \quad 4 \quad \exists a \in \mathbb{N} \exists b \in \mathbb{N} x = (a, b) & \text{Th. 3.16.5.4}(\wedge E1(3)) \\ 5 \quad 5 \quad a, b \in \mathbb{N} \wedge x = (a, b) & A \\ 3,5 \quad 6 \quad z = [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & = E(5, \wedge E2(3)) \\ 5 \quad 7 \quad [a, b]_{\mathbb{Z}} = [(a, b)]_{\sim_{\mathbb{Z}}, D_{\mathbb{Z}}} & \text{Def. 13.2.2.2(5)} \\ 3,5 \quad 8 \quad z = [a, b]_{\mathbb{Z}} & \text{Th. 2.9.1.3(6,7)} \\ 3 \quad 9 \quad \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} & \exists E(4, 5, 8) \\ 1 \quad 10 \quad \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} & \exists E(2, 3, 9) \end{array}$$

□

2.3 Einbettung der natürlichen Zahlen

Definition 13.2.3.1 (Natürliche Einbettung).

Mengen: $\mathbb{N}$
Funktionensymbol: $\iota_{\mathbb{Z}}$
Neue Symbole: $\iota_{\mathbb{Z}}$

$$\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z}, \quad n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(n) := [n, 0]_{\mathbb{Z}}$$

$$\begin{array}{ll} 1 & 1 \quad n \in \mathbb{N} \quad A \\ 2 & 0 \in \mathbb{N} \quad Ax. \ 12.3.1 \\ 1 & 3 \quad [n, 0]_{\mathbb{Z}} \in \mathbb{Z} \quad Th. \ 13.2.2.1(1, 2) \end{array}$$

Theorem 13.2.3.1 (Typisierung der natürlichen Einbettung).

Funktion: $\iota_{\mathbb{Z}}$

$$\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z}$$

$$1 \quad \iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z} \quad Def. \ 13.2.3.1$$

Theorem 13.2.3.2 (Injektivität der natürlichen Einbettung).

Mengen: m, n
Funktion: $\iota_{\mathbb{Z}}$

$$\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z}$$

Beweis.

Gleichheit von eingebetteten Zahlen

Th. 13.2.3.2(i)

$$m, n \in \mathbb{N}, \quad \iota_{\mathbb{Z}}(m) = \iota_{\mathbb{Z}}(n) \vdash m = n$$

$$\begin{array}{ll} 1 & 1 \quad m, n \in \mathbb{N} \quad A \\ 2 & 2 \quad \iota_{\mathbb{Z}}(m) = \iota_{\mathbb{Z}}(n) \quad A \\ 1 & 3 \quad \iota_{\mathbb{Z}}(m) = [m, 0]_{\mathbb{Z}} \quad Def. \ 13.2.3.1(1) \\ 1 & 4 \quad \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}} \quad Def. \ 13.2.3.1(1) \\ 1,2 & 5 \quad [m, 0]_{\mathbb{Z}} = [n, 0]_{\mathbb{Z}} = E(3, 4, 2) \\ 1,2 & 6 \quad m + 0 = 0 + n \quad Th. \ 13.2.2.2(1,5) \\ 1 & 7 \quad m + 0 = m \quad Th. \ 12.4.4.2(1) \\ 1 & 8 \quad 0 + n = n \quad Th. \ 12.4.4.4(1) \end{array}$$

$$1,2 \quad 9 \quad m = n \quad = E(7, 8, 6)$$

Schluss:

$$\begin{array}{ll} 1 \quad \iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z} & \text{Th. 13.2.3.1} \\ 2 \quad 2 \quad m \in \mathbb{N} & A \\ 3 \quad 3 \quad n \in \mathbb{N} & A \\ 4 \quad 4 \quad \iota_{\mathbb{Z}}(m) = \iota_{\mathbb{Z}}(n)A & \\ 2,3,4 \quad 5 \quad m = n & \text{Th. 13.2.3.2(i)(2,3,4)} \\ 6 \quad \iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z} & \text{Def. 6.2.1.1}(1, \forall I(\rightarrow I(2, 3, 4, 5))) \end{array}$$

□

Definition 13.2.3.2 (Null der ganzen Zahlen).

Mengen: $0_{\mathbb{Z}}$
Neue Symbole: $0_{\mathbb{Z}}$

$$0_{\mathbb{Z}} := \iota_{\mathbb{Z}}(0)$$

Definition 13.2.3.3 (Eins der ganzen Zahlen).

Mengen: $1_{\mathbb{Z}}$
Neue Symbole: $1_{\mathbb{Z}}$

$$1_{\mathbb{Z}} := \iota_{\mathbb{Z}}(1)$$

Theorem 13.2.3.3 (Nullklasse).

Mengen: $0_{\mathbb{Z}}$

$$0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}$$

$$\begin{array}{ll} 1 \quad 0_{\mathbb{Z}} = \iota_{\mathbb{Z}}(0) & \text{Def. 13.2.3.2} \\ 2 \quad 0 \in \mathbb{N} & \text{Ax. 12.3.1} \\ 3 \quad \iota_{\mathbb{Z}}(0) = [0, 0]_{\mathbb{Z}} & \text{Def. 13.2.3.1(2)} \\ 4 \quad 0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}} & \text{Th. 2.9.1.2(1, 3)} \end{array}$$

Theorem 13.2.3.4 (Die Ganzzahlnull liegt im Quotienten).

Mengen: $0_{\mathbb{Z}}$

$$0_{\mathbb{Z}} \in \mathbb{Z}$$

- 1 $0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}$ Th. 13.2.3.3
- 2 $0 \in \mathbb{N}$ Ax. 12.3.1
- 3 $[0, 0]_{\mathbb{Z}} \in \mathbb{Z}$ Th. 13.2.2.1(2, 2)
- 4 $0_{\mathbb{Z}} \in \mathbb{Z} = E(1, 3)$

Theorem 13.2.3.5 (Die Ganzzahleins liegt im Quotienten).

Mengen: $1_{\mathbb{Z}}$

$$1_{\mathbb{Z}} \in \mathbb{Z}$$

- 1 $1_{\mathbb{Z}} = \iota_{\mathbb{Z}}(1)$ Def. 13.2.3.3
- 2 $1 \in \mathbb{N}$ Th. 12.4.4.11
- 3 $\iota_{\mathbb{Z}}: \mathbb{N} \rightarrow \mathbb{Z}$ Th. 13.2.3.1
- 4 $\iota_{\mathbb{Z}}(1) \in \mathbb{Z}$ Th. 5.2.3.10(3, 2)
- 5 $1_{\mathbb{Z}} \in \mathbb{Z} = E(1, 4)$

2.4 Negation und Addition

2.4.1 Negation

Theorem 13.2.4.1 (Repräsentantenunabhängigkeit der Negation).

Mengen: a, b, c, d

$$a, b, c, d \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \vdash \\ [b, a]_{\mathbb{Z}} = [d, c]_{\mathbb{Z}}$$

Beweis.

Kreuzsummen des Ausgangspaares

Th. 13.2.4.1(i)

$$a, b, c, d \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \vdash a + d = b + c$$

- 1 $1 \ a, b, c, d \in \mathbb{N} \quad A$
- 2 $2 \ [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \quad A$
- 1,2 $3 \ a + d = b + c \quad \text{Th. 13.2.2.2(1,2)}$

Vertauschen der Koordinaten:

- 1 $1 \ a, b, c, d \in \mathbb{N} \quad A$

$$\begin{array}{ll}
2 & 2 \ [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} A \\
1,2 & 3 \ a + d = b + c \quad \text{Th. 13.2.4.1(i)(1,2)} \\
1,2 & 4 \ b + c = a + d \quad \text{Th. 2.9.1.1(3)} \\
1,2 & 5 \ [b, a]_{\mathbb{Z}} = [d, c]_{\mathbb{Z}} \text{Th. 13.2.2.2(1,4)}
\end{array}$$

□

Theorem 13.2.4.2 (Quotientenabstieg der Negation).

Mengen: $z, u, v, a, b, c, d, p, q$

$$z \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}})$$

Beweis.

Existenz aus einem Repräsentanten

Th. 13.2.4.2(i)

$$z \in \mathbb{Z} \vdash \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}})$$

$$\begin{array}{ll}
1 & 1 \ z \in \mathbb{Z} \quad A \\
1 & 2 \ \exists a \in \mathbb{N} \exists b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}} \quad \text{Th. 13.2.2.3(1)} \\
3 & 3 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \quad A \\
3 & 4 \ [b, a]_{\mathbb{Z}} \in \mathbb{Z} \quad \text{Th. 13.2.2.1(3)} \\
5 & 5 \ [b, a]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}} \quad = I \\
3 & 6 \ \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge [b, a]_{\mathbb{Z}} = [q, p]_{\mathbb{Z}}) \quad \exists I(\wedge I(3, 5)) \\
3 & 7 \ [b, a]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge [b, a]_{\mathbb{Z}} = [q, p]_{\mathbb{Z}}) \wedge I(4, 6) \\
3 & 8 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}}) \quad \exists I(7) \\
1 & 9 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}}) \quad \exists E(2, 3, 8)
\end{array}$$

Eindeutigkeit für feste Repräsentanten

Th. 13.2.4.2(ii)

$$\begin{array}{l}
a, b, c, d \in \mathbb{N}, \ z = [a, b]_{\mathbb{Z}}, \ u = [b, a]_{\mathbb{Z}}, \\
z = [c, d]_{\mathbb{Z}}, \ v = [d, c]_{\mathbb{Z}} \vdash u = v
\end{array}$$

$$\begin{array}{ll}
1 & 1 \ a, b, c, d \in \mathbb{N} \quad A \\
2 & 2 \ z = [a, b]_{\mathbb{Z}} \quad A \\
3 & 3 \ u = [b, a]_{\mathbb{Z}} \quad A \\
4 & 4 \ z = [c, d]_{\mathbb{Z}} \quad A \\
5 & 5 \ v = [d, c]_{\mathbb{Z}} \quad A \\
2,4 & 6 \ [a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} = E(2, 4) \\
1,2,4 & 7 \ [b, a]_{\mathbb{Z}} = [d, c]_{\mathbb{Z}} \text{Th. 13.2.4.1(1,6)} \\
1,2,3,4,5 & 8 \ u = v \quad = E(3, 5, 7)
\end{array}$$

Eindeutigkeit beliebiger Ergebniszeugen

Th. 13.2.4.2(iii)

$$\begin{aligned} & (u \in \mathbb{Z} \wedge \exists a \in \mathbb{N} \exists b \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge u = [b, a]_{\mathbb{Z}})), \\ & (v \in \mathbb{Z} \wedge \exists c \in \mathbb{N} \exists d \in \mathbb{N} (z = [c, d]_{\mathbb{Z}} \wedge v = [d, c]_{\mathbb{Z}})) \vdash u = v \end{aligned}$$

1	$1 \ u \in \mathbb{Z} \wedge \exists a \in \mathbb{N} \exists b \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge u = [b, a]_{\mathbb{Z}}) A$	
2	$2 \ v \in \mathbb{Z} \wedge \exists c \in \mathbb{N} \exists d \in \mathbb{N} (z = [c, d]_{\mathbb{Z}} \wedge v = [d, c]_{\mathbb{Z}}) A$	
1	$3 \ \exists a \in \mathbb{N} \exists b \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge u = [b, a]_{\mathbb{Z}})$	$\wedge E2(1)$
2	$4 \ \exists c \in \mathbb{N} \exists d \in \mathbb{N} (z = [c, d]_{\mathbb{Z}} \wedge v = [d, c]_{\mathbb{Z}})$	$\wedge E2(2)$
5	$5 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge u = [b, a]_{\mathbb{Z}}$	A
6	$6 \ c, d \in \mathbb{N} \wedge z = [c, d]_{\mathbb{Z}} \wedge v = [d, c]_{\mathbb{Z}}$	A
5	$7 \ a, b \in \mathbb{N}$	$\wedge E1(5)$
5	$8 \ z = [a, b]_{\mathbb{Z}}$	$\wedge E1(\wedge E2(5))$
5	$9 \ u = [b, a]_{\mathbb{Z}}$	$\wedge E2(\wedge E2(5))$
6	$10 \ c, d \in \mathbb{N}$	$\wedge E1(6)$
6	$11 \ z = [c, d]_{\mathbb{Z}}$	$\wedge E1(\wedge E2(6))$
6	$12 \ v = [d, c]_{\mathbb{Z}}$	$\wedge E2(\wedge E2(6))$
5,6	$13 \ a, b, c, d \in \mathbb{N}$	$\wedge I(7, 10)$
5,6	$14 \ u = v$	Th. 13.2.4.2(ii)(13,8,9,11,12)
5	$15 \ u = v$	$\exists E(4, 6, 14)$
1,2	$16 \ u = v$	$\exists E(3, 5, 15)$

Abschluss des Abstiegs

Th. 13.2.4.2(iv)

$$z \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}})$$

1	$1 \ z \in \mathbb{Z}$	A
1	$2 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}})$	Th. 13.2.4.2(i)(1)
3	$3 \ u \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}}) A$	
4	$4 \ v \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge v = [q, p]_{\mathbb{Z}}) A$	
3,4	$5 \ u = v$	Th. 13.2.4.2(iii)(3,4)
1	$6 \ \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge u = [q, p]_{\mathbb{Z}})$	$\exists! I(2, 3, 4, 5)$

□

Der vorangehende Satz zeigt, dass die folgende Klassenregel unabhängig vom gewählten Differenzenpaar genau ein Ergebnis in $\mathbb{Z}$ festlegt.

Definition 13.2.4.1 (Negation ganzer Zahlen).

Mengen: a, b
Neue Symbole: $-$

$$a, b \in \mathbb{N} \vdash -[a, b]_{\mathbb{Z}} := [b, a]_{\mathbb{Z}}$$

Theorem 13.2.4.3 (Abgeschlossenheit der Negation).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash -z \in \mathbb{Z}$$

1	1	$z \in \mathbb{Z}$	A
1	2	$\exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	$Th. 13.2.2.3(1)$
3	3	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	4	$-[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}}$	$Def. 13.2.4.1(3)$
3	5	$[b, a]_{\mathbb{Z}} \in \mathbb{Z}$	$Th. 13.2.2.1(3)$
3	6	$-z \in \mathbb{Z}$	$= E(3, 4, 5)$
1	7	$-z \in \mathbb{Z}$	$\exists E(2, 3, 6)$

Theorem 13.2.4.4 (Doppelte Negation).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash -(-z) = z$$

1	1	$z \in \mathbb{Z}$	A
1	2	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	$Th. 13.2.2.3(1)$
3	3	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	4	$-[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}}$	$Def. 13.2.4.1(3)$
3	5	$-[b, a]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}}$	$Def. 13.2.4.1(3)$
3	6	$-(-z) = z$	$= E(3, 4, 5)$
1	7	$-(-z) = z$	$\exists E(2, 3, 6)$

2.4.2 Addition

Theorem 13.2.4.5 (Repräsentantenunabhängigkeit der Addition).

Mengen: $a, b, a', b', c, d, c', d'$

$$a, b, a', b', c, d, c', d' \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}, [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ [a + c, b + d]_{\mathbb{Z}} = [a' + c', b' + d']_{\mathbb{Z}}$$

Beweis.

Kreuzsummen der beiden Eingaben

Th. 13.2.4.5(i)

$$a, b, a', b', c, d, c', d' \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}, [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ a + b' = b + a' \quad \wedge \quad c + d' = d + c'$$

$$\begin{array}{ll} 1 & 1 \ a, b, a', b', c, d, c', d' \in \mathbb{N} \quad A \\ 2 & 2 \ [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}} \quad A \\ 3 & 3 \ [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \quad A \\ 1,2 & 4 \ a + b' = b + a' \quad \text{Th. 13.2.2.2(1,2)} \\ 1,3 & 5 \ c + d' = d + c' \quad \text{Th. 13.2.2.2(1,3)} \\ 1,2,3 & 6 \ a + b' = b + a' \wedge c + d' = d + c' \wedge I(4,5) \end{array}$$

Zusammenführen und Umsortieren

Th. 13.2.4.5(ii)

$$a, b, a', b', c, d, c', d' \in \mathbb{N}, a + b' = b + a', c + d' = d + c' \vdash \\ (a + c) + (b' + d') = (b + d) + (a' + c')$$

$$\begin{array}{ll} 1 & 1 \ a, b, a', b', c, d, c', d' \in \mathbb{N} \quad A \\ 2 & 2 \ a + b' = b + a' \quad A \\ 3 & 3 \ c + d' = d + c' \quad A \\ 1,2,3 & 4 \ (a + b') + (c + d') = (b + a') + (d + c') = E(2,3) \\ 1 & 5 \ (a + b') + (c + d') = (a + c) + (b' + d') \text{Th. 12.4.4.10(1)} \\ 1 & 6 \ (b + a') + (d + c') = (b + d) + (a' + c') \text{Th. 12.4.4.10(1)} \\ 1,2,3 & 7 \ (a + c) + (b' + d') = (b + d) + (a' + c') = E(4,5,6) \end{array}$$

Schluss:

$$\begin{array}{ll} 1 & 1 \ a, b, a', b', c, d, c', d' \in \mathbb{N} \quad A \\ 2 & 2 \ [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}} \quad A \\ 3 & 3 \ [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \quad A \\ 1,2,3 & 4 \ a + b' = b + a' \wedge c + d' = d + c' \quad \text{Th. 13.2.4.5(i)(1,2,3)} \\ 1,2,3 & 5 \ (a + c) + (b' + d') = (b + d) + (a' + c') \text{Th. 13.2.4.5(ii)(1, \wedge E1(4), \wedge E2(4))} \\ 1,2,3 & 6 \ [a + c, b + d]_{\mathbb{Z}} = [a' + c', b' + d']_{\mathbb{Z}} \quad \text{Th. 13.2.2.2(1,5)} \end{array}$$

□

Theorem 13.2.4.6 (Quotientenabstieg der Addition).

Mengen: $z, w, u, v, a, b, c, d, a', b', c', d', p, q, r, s$

$$z, w \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})$$

Beweis.

Ergebniskandidat fester Repräsentanten

Th. 13.2.4.6(i)

$$\begin{aligned} a, b, c, d \in \mathbb{N}, \quad z = [a, b]_{\mathbb{Z}}, \quad w = [c, d]_{\mathbb{Z}} \vdash \\ [a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [a + c, b + d]_{\mathbb{Z}} = [p + r, q + s]_{\mathbb{Z}}) \end{aligned}$$

$$\begin{aligned} 1 \quad & 1 \ a, b, c, d \in \mathbb{N} \\ 2 \quad & 2 \ z = [a, b]_{\mathbb{Z}} \\ 3 \quad & 3 \ w = [c, d]_{\mathbb{Z}} \\ 1 \quad & 4 \ a + c \in \mathbb{N} \\ 1 \quad & 5 \ b + d \in \mathbb{N} \\ 1 \quad & 6 \ [a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \\ & 7 \ [a + c, b + d]_{\mathbb{Z}} = [a + c, b + d]_{\mathbb{Z}} \\ 1,2,3 \quad & 8 \ \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [a + c, b + d]_{\mathbb{Z}} = [p + r, q + s]_{\mathbb{Z}}) \\ 1,2,3 \quad & 9 \ [a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [a + c, b + d]_{\mathbb{Z}} = [p + r, q + s]_{\mathbb{Z}}) \end{aligned}$$

Existenz durch getrennte Repräsentantenwahl

Th. 13.2.4.6(ii)

$$\begin{aligned} z, w \in \mathbb{Z} \vdash \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) \end{aligned}$$

$$\begin{aligned} 1 \quad & 1 \ z \in \mathbb{Z} \\ 2 \quad & 2 \ w \in \mathbb{Z} \\ 1 \quad & 3 \ \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} \\ 2 \quad & 4 \ \exists c \in \mathbb{N} \exists d \in \mathbb{N} w = [c, d]_{\mathbb{Z}} \\ 5 \quad & 5 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \\ 6 \quad & 6 \ c, d \in \mathbb{N} \wedge w = [c, d]_{\mathbb{Z}} \\ 5 \quad & 7 \ a, b \in \mathbb{N} \\ 5 \quad & 8 \ z = [a, b]_{\mathbb{Z}} \\ 6 \quad & 9 \ c, d \in \mathbb{N} \\ 6 \quad & 10 \ w = [c, d]_{\mathbb{Z}} \\ 5,6 \quad & 11 \ a, b, c, d \in \mathbb{N} \\ 5,6 \quad & 12 \ [a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [a + c, b + d]_{\mathbb{Z}} = [p + r, q + s]_{\mathbb{Z}}) \\ 5,6 \quad & 13 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) \\ & 5 \ 14 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) \\ 1,2 \quad & 15 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) \end{aligned}$$

Eindeutigkeit für feste Repräsentanten

Th. 13.2.4.6(iii)

$$\begin{aligned}
& a, b, c, d, a', b', c', d' \in \mathbb{N}, \\
& z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}}, \\
& z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}} \vdash u = v
\end{aligned}$$

1	$1 \ a, b, c, d, a', b', c', d' \in \mathbb{N}$	A
2	$2 \ z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}}$	A
3	$3 \ z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}}$	A
2	$4 \ z = [a, b]_{\mathbb{Z}}$	$\wedge E1(2)$
2	$5 \ w = [c, d]_{\mathbb{Z}}$	$\wedge E1(\wedge E2(2))$
2	$6 \ u = [a + c, b + d]_{\mathbb{Z}}$	$\wedge E2(\wedge E2(2))$
3	$7 \ z = [a', b']_{\mathbb{Z}}$	$\wedge E1(3)$
3	$8 \ w = [c', d']_{\mathbb{Z}}$	$\wedge E1(\wedge E2(3))$
3	$9 \ v = [a' + c', b' + d']_{\mathbb{Z}}$	$\wedge E2(\wedge E2(3))$
2,3	$10 \ [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}$	$= E(4, 7)$
2,3	$11 \ [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}}$	$= E(5, 8)$
1,2,3	$12 \ [a + c, b + d]_{\mathbb{Z}} = [a' + c', b' + d']_{\mathbb{Z}}$	$\text{Th. 13.2.4.5}(1, 10, 11)$
1,2,3	$13 \ u = v$	$= E(6, 9, 12)$

Eindeutigkeit beliebiger Ergebniszeugen

Th. 13.2.4.6(iv)

$$\begin{aligned}
& (u \in \mathbb{Z} \wedge \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}})), \\
& (v \in \mathbb{Z} \wedge \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}})) \vdash u = v
\end{aligned}$$

1	$1 \ u \in \mathbb{Z} \wedge \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}})$	A
2	$2 \ v \in \mathbb{Z} \wedge \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}})$	A
1	$3 \ \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}})$	$\wedge E2(1)$
2	$4 \ \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}})$	$\wedge E2(2)$
5	$5 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}}$	A
6	$6 \ a', b', c', d' \in \mathbb{N} \wedge z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}}$	A
5	$7 \ a, b, c, d \in \mathbb{N}$	$\wedge E1(5)$
5	$8 \ z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [a + c, b + d]_{\mathbb{Z}}$	$\wedge E2(5)$
6	$9 \ a', b', c', d' \in \mathbb{N}$	$\wedge E1(6)$
6	$10 \ z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a' + c', b' + d']_{\mathbb{Z}}$	$\wedge E2(6)$
5,6	$11 \ a, b, c, d, a', b', c', d' \in \mathbb{N}$	$\wedge I(7, 9)$
5,6	$12 \ u = v$	$\text{Th. 13.2.4.6(iii)}(11, 8, 10)$
5	$13 \ u = v$	$\exists E(4, 6, 12)$
1,2	$14 \ u = v$	$\exists E(3, 5, 13)$

Abschluss des Abstiegs

Th. 13.2.4.6(v)

$$\begin{aligned}
& z, w \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\
& (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}})
\end{aligned}$$

$$\begin{array}{ll}
1 & 1 \ z, w \in \mathbb{Z} \quad A \\
1 & 2 \ \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) \text{ Th. 13.2.4.6(ii)(1)} \\
3 & 3 \ u \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) A \\
4 & 4 \ v \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge v = [p + r, q + s]_{\mathbb{Z}}) A \\
3,4 & 5 \ u = v \quad \text{Th. 13.2.4.6(iv)(3,4)} \\
1 & 6 \ \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [p + r, q + s]_{\mathbb{Z}}) \exists! I(2, 3, 4, 5)
\end{array}$$

□

Der eindeutige Existenzsatz ist der Quotientenabstieg der Addition; die folgende Klassenregel bezeichnet dieses eindeutig bestimmte Ergebnis.

Definition 13.2.4.2 (Addition ganzer Zahlen).

Mengen: a, b, c, d
Neue Symbole: $+$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} := [a + c, b + d]_{\mathbb{Z}}$$

Theorem 13.2.4.7 (Abgeschlossenheit der Addition ganzer Zahlen).

Mengen: z, w, a, b, c, d

$$z \in \mathbb{Z}, w \in \mathbb{Z} \vdash z + w \in \mathbb{Z}$$

Beweis.

Abgeschlossenheit für feste Repräsentanten Th. 13.2.4.7(i)

$$\begin{array}{l}
a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \\
\vdash z + w \in \mathbb{Z}
\end{array}$$

$$\begin{array}{ll}
1 & 1 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A \\
1 & 2 \ z + w = [a + c, b + d]_{\mathbb{Z}} \quad \text{Def. 13.2.4.2(1)} \\
1 & 3 \ a + c \in \mathbb{N} \quad \text{Th. 12.4.4.1(1)} \\
1 & 4 \ b + d \in \mathbb{N} \quad \text{Th. 12.4.4.1(1)} \\
1 & 5 \ [a + c, b + d]_{\mathbb{Z}} \in \mathbb{Z} \quad \text{Th. 13.2.2.1(3,4)} \\
1 & 6 \ z + w \in \mathbb{Z} \quad = E(2, 5)
\end{array}$$

Elimination der Repräsentanten Th. 13.2.4.7(ii)

$$z \in \mathbb{Z}, w \in \mathbb{Z} \vdash z + w \in \mathbb{Z}$$

$$1 \ 1 \ z \in \mathbb{Z} \quad A$$

2	$2 w \in \mathbb{Z}$	A
1	$3 \exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(1)
2	$4 \exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.3(2)
5	$5 a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}}$	A
5	$6 z + w \in \mathbb{Z}$	Th. 13.2.4.7(i)(5)
1,2	$7 z + w \in \mathbb{Z}$	$\exists E(3, 4, 5, 6)$

□

Theorem 13.2.4.8 (Assoziativität der Addition ganzer Zahlen).

Mengen: $z, w, u, a, b, c, d, e, f$

$$z, w, u \in \mathbb{Z} \vdash (z + w) + u = z + (w + u)$$

Beweis.

Rechnung mit Repräsentanten

Th. 13.2.4.8(i)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$([a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}}) + [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} + ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}})$$

1	$1 a, b, c, d, e, f \in \mathbb{N}$	A
1	$2 ([a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}}) + [e, f]_{\mathbb{Z}} = [(a + c) + e, (b + d) + f]_{\mathbb{Z}}$	Def. 13.2.4.2(1)
1	$3 [a, b]_{\mathbb{Z}} + ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a + (c + e), b + (d + f)]_{\mathbb{Z}}$	Def. 13.2.4.2(1)
1	$4 (a + c) + e = a + (c + e)$	Th. 12.4.4.7(1)
1	$5 (b + d) + f = b + (d + f)$	Th. 12.4.4.7(1)
1	$6 [(a + c) + e, (b + d) + f]_{\mathbb{Z}} = [a + (c + e), b + (d + f)]_{\mathbb{Z}} = E(4, 5)$	
1	$7 ([a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}}) + [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} + ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = E(2, 3, 6)$	

Schluss:

1	$1 z, w, u \in \mathbb{Z}$	A
1	$2 z \in \mathbb{Z}$	$\wedge E1(1)$
1	$3 w \in \mathbb{Z}$	$\wedge E1(\wedge E2(1))$
1	$4 u \in \mathbb{Z}$	$\wedge E2(\wedge E2(1))$
1	$5 \exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(2)
1	$6 \exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.3(3)
1	$7 \exists e, f \in \mathbb{N} u = [e, f]_{\mathbb{Z}}$	Th. 13.2.2.3(4)
8	$8 a, b, c, d, e, f \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [e, f]_{\mathbb{Z}}$	A
8	$9 ([a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}}) + [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} + ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}})$	Th. 13.2.4.8(i)(8)
8	$10 (z + w) + u = z + (w + u)$	$= E(8, 9)$
1	$11 (z + w) + u = z + (w + u)$	$\exists E(5, 6, 7, 8, 10)$

□

Theorem 13.2.4.9 (Kommutativität der Addition ganzer Zahlen).

Mengen: z, w, a, b, c, d

$$z, w \in \mathbb{Z} \vdash z + w = w + z$$

Beweis.

Vertauschung fester Repräsentanten

Th. 13.2.4.9(i)

$$\begin{aligned} a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \\ \vdash z + w = w + z \end{aligned}$$

$$\begin{array}{ll} 1 \quad 1 \quad a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} & A \\ 1 \quad 2 \quad z + w = [a + c, b + d]_{\mathbb{Z}} & \text{Def. 13.2.4.2(1)} \\ 1 \quad 3 \quad w + z = [c + a, d + b]_{\mathbb{Z}} & \text{Def. 13.2.4.2(1)} \\ 1 \quad 4 \quad a + c = c + a & \text{Th. 12.4.4.8(1)} \\ 1 \quad 5 \quad b + d = d + b & \text{Th. 12.4.4.8(1)} \\ 1 \quad 6 \quad [a + c, b + d]_{\mathbb{Z}} = [c + a, d + b]_{\mathbb{Z}} & = E(4, 5) \\ 1 \quad 7 \quad z + w = w + z & = E(2, 3, 6) \end{array}$$

Getrennte Repräsentantenwahl

Th. 13.2.4.9(ii)

$$z, w \in \mathbb{Z} \vdash z + w = w + z$$

$$\begin{array}{ll} 1 \quad 1 \quad z, w \in \mathbb{Z} & A \\ 1 \quad 2 \quad z \in \mathbb{Z} & \wedge E1(1) \\ 1 \quad 3 \quad w \in \mathbb{Z} & \wedge E2(1) \\ 1 \quad 4 \quad \exists a, b \in \mathbb{N} \quad z = [a, b]_{\mathbb{Z}} & \text{Th. 13.2.2.3(2)} \\ 1 \quad 5 \quad \exists c, d \in \mathbb{N} \quad w = [c, d]_{\mathbb{Z}} & \text{Th. 13.2.2.3(3)} \\ 6 \quad 6 \quad a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} & A \\ 6 \quad 7 \quad z + w = w + z & \text{Th. 13.2.4.9(i)(6)} \\ 1 \quad 8 \quad z + w = w + z & \exists E(4, 5, 6, 7) \end{array}$$

□

Theorem 13.2.4.10 (Nullgesetze der Addition ganzer Zahlen).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash z + 0_{\mathbb{Z}} = z \wedge 0_{\mathbb{Z}} + z = z$$

Beweis.

Rechtsneutrales Element für einen Repräsentanten Th. 13.2.4.10(i)

$$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \\ \vdash z + 0_{\mathbb{Z}} = z$$

$$\begin{array}{ll} 1 & 1 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \quad A \\ & 2 \ 0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}} \quad \text{Th. 13.2.3.3} \\ 1 & 3 \ z + 0_{\mathbb{Z}} = [a + 0, b + 0]_{\mathbb{Z}} \text{Def. 13.2.4.2(1,2)} \\ 1 & 4 \ a + 0 = a \quad \text{Th. 12.4.4.2(1)} \\ 1 & 5 \ b + 0 = b \quad \text{Th. 12.4.4.2(1)} \\ 1 & 6 \ z + 0_{\mathbb{Z}} = z \quad = E(1, 3, 4, 5) \end{array}$$

Elimination des Repräsentanten

Th. 13.2.4.10(ii)

$$z \in \mathbb{Z} \vdash z + 0_{\mathbb{Z}} = z \wedge 0_{\mathbb{Z}} + z = z$$

$$\begin{array}{ll} 1 & 1 \ z \in \mathbb{Z} \quad A \\ 1 & 2 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}} \quad \text{Th. 13.2.2.3(1)} \\ 3 & 3 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \quad A \\ 3 & 4 \ z + 0_{\mathbb{Z}} = z \quad \text{Th. 13.2.4.10(i)(3)} \\ 3 & 5 \ 0_{\mathbb{Z}} + z = z \quad \text{Th. 13.2.4.9(Th. 13.2.3.4,1, Th. 2.9.1.1(4))} \\ 3 & 6 \ z + 0_{\mathbb{Z}} = z \wedge 0_{\mathbb{Z}} + z = z \wedge I(4, 5) \\ 1 & 7 \ z + 0_{\mathbb{Z}} = z \wedge 0_{\mathbb{Z}} + z = z \exists E(2, 3, 6) \end{array}$$

□

Theorem 13.2.4.11 (Additive Inversenregeln).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash z + (-z) = 0_{\mathbb{Z}} \wedge (-z) + z = 0_{\mathbb{Z}}$$

Beweis.

Inverse Rechnung auf einer Klasse

Th. 13.2.4.11(i)

$$a, b \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} + (-[a, b]_{\mathbb{Z}}) = 0_{\mathbb{Z}}$$

$$\begin{array}{ll} 1 & 1 \ a, b \in \mathbb{N} \quad A \\ 1 & 2 \ -[a, b]_{\mathbb{Z}} = [b, a]_{\mathbb{Z}} \quad \text{Def. 13.2.4.1(1)} \\ 1 & 3 \ [a, b]_{\mathbb{Z}} + (-[a, b]_{\mathbb{Z}}) = [a + b, b + a]_{\mathbb{Z}} \text{Def. 13.2.4.2(1,2)} \end{array}$$

1 4	$a + b = b + a$	Th. 12.4.4.8(1)
1 5	$[a + b, b + a]_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}$	Th. 13.2.2.2(Th. 12.4.4.1(1), Th. 12.4.4.1(1), Ax. 12.3.1, Ax. 12.3.1, = I(3, 5, 6))
6	$0_{\mathbb{Z}} = [0, 0]_{\mathbb{Z}}$	Th. 13.2.3.3
1 7	$[a, b]_{\mathbb{Z}} + (-[a, b]_{\mathbb{Z}}) = 0_{\mathbb{Z}}$	= E(3, 5, 6)

Schluss:

1 1	$z \in \mathbb{Z}$	A
1 2	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(1)
3 3	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3 4	$z + (-z) = 0_{\mathbb{Z}}$	Th. 13.2.4.11(i)(3)
1 5	$-z \in \mathbb{Z}$	Th. 13.2.4.3(1)
1,3 6	$(-z) + z = z + (-z)$	Th. 13.2.4.9(5,1)
1,3 7	$(-z) + z = 0_{\mathbb{Z}}$	= E(6, 4)
1,3 8	$z + (-z) = 0_{\mathbb{Z}} \wedge (-z) + z = 0_{\mathbb{Z}} \wedge I(4, 7)$	
1 9	$z + (-z) = 0_{\mathbb{Z}} \wedge (-z) + z = 0_{\mathbb{Z}} \exists E(2, 3, 8)$	

□

Theorem 13.2.4.12 (Negation der Ganzzahlnull).

Mengen: $0_{\mathbb{Z}}$

$$-0_{\mathbb{Z}} = 0_{\mathbb{Z}}$$

1	$0_{\mathbb{Z}} \in \mathbb{Z}$	Th. 13.2.3.4
2	$-0_{\mathbb{Z}} \in \mathbb{Z}$	Th. 13.2.4.3(1)
3	$(-0_{\mathbb{Z}}) + 0_{\mathbb{Z}} = -0_{\mathbb{Z}} \wedge E1(\text{Th. 13.2.4.10}(2))$	
4	$(-0_{\mathbb{Z}}) + 0_{\mathbb{Z}} = 0_{\mathbb{Z}} \wedge E2(\text{Th. 13.2.4.11}(1))$	
5	$-0_{\mathbb{Z}} = (-0_{\mathbb{Z}}) + 0_{\mathbb{Z}}$	Th. 2.9.1.1(3)
6	$-0_{\mathbb{Z}} = 0_{\mathbb{Z}}$	= E(5, 4)

2.5 Multiplikation

Theorem 13.2.5.1 (Repräsentantenunabhängigkeit der Multiplikation).

Mengen: $a, b, a', b', c, d, c', d'$

$$a, b, a', b', c, d, c', d' \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}, [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ [a \cdot c + b \cdot d, a \cdot d + b \cdot c]_{\mathbb{Z}} = [a' \cdot c' + b' \cdot d', a' \cdot d' + b' \cdot c']_{\mathbb{Z}}$$

Beweis.

Wechsel des ersten Repräsentanten

Th. 13.2.5.1(i)

$$a, b, a', b', c, d \in \mathbb{N}, [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}} \vdash \\ [ac + bd, ad + bc]_{\mathbb{Z}} = [a'c + b'd, a'd + b'c]_{\mathbb{Z}}$$

1	1	$a, b, a', b', c, d \in \mathbb{N}$	A
2	2	$[a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}$	A
1,2	3	$a + b' = b + a'$	Th. 13.2.2.2(1,2)
1,2	4	$(a + b')c = (b + a')c$	$= E(3)$
1	5	$(a + b')c = ac + b'c$	Th. 12.4.7.12(1)
1	6	$(b + a')c = bc + a'c$	Th. 12.4.7.12(1)
1,2	7	$ac + b'c = bc + a'c$	$= E(4, 5, 6)$
1,2	8	$(a + b')d = (b + a')d$	$= E(3)$
1	9	$(a + b')d = ad + b'd$	Th. 12.4.7.12(1)
1	10	$(b + a')d = bd + a'd$	Th. 12.4.7.12(1)
1,2	11	$ad + b'd = bd + a'd$	$= E(8, 9, 10)$
1	12	$a'd + b'c = b'c + a'd$	Th. 12.4.4.8(1)
1,2	13	$(ac + bd) + (a'd + b'c) = (ac + bd) + (b'c + a'd)$	$= E(12)$
1	14	$(ac + bd) + (b'c + a'd) = (ac + b'c) + (bd + a'd)$	Th. 12.4.4.10(1)
1,2	15	$bd + a'd = ad + b'd$	Th. 2.9.1.1(11)
1,2	16	$(ac + b'c) + (bd + a'd) = (bc + a'c) + (ad + b'd)$	$= E(7, 15)$
1	17	$(bc + a'c) + (ad + b'd) = (bc + ad) + (a'c + b'd)$	Th. 12.4.4.10(1)
1	18	$bc + ad = ad + bc$	Th. 12.4.4.8(1)
1	19	$(bc + ad) + (a'c + b'd) = (ad + bc) + (a'c + b'd)$	$= E(18)$
1,2	20	$(ac + bd) + (a'd + b'c) = (ad + bc) + (a'c + b'd)$	$= E(13, 14, 16, 17, 19)$
1,2	21	$[ac + bd, ad + bc]_{\mathbb{Z}} = [a'c + b'd, a'd + b'c]_{\mathbb{Z}}$	Th. 13.2.2.2(1,20)

Wechsel des zweiten Repräsentanten

Th. 13.2.5.1(ii)

$$a, b, c, d, c', d' \in \mathbb{N}, [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ [ac + bd, ad + bc]_{\mathbb{Z}} = [ac' + bd', ad' + bc']_{\mathbb{Z}}$$

1	1	$a, b, c, d, c', d' \in \mathbb{N}$	A
2	2	$[c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}}$	A
1,2	3	$c + d' = d + c'$	Th. 13.2.2.2(1,2)
1,2	4	$a(c + d') = a(d + c')$	$= E(3)$
1	5	$a(c + d') = ac + ad'$	Th. 12.4.7.11(1)
1	6	$a(d + c') = ad + ac'$	Th. 12.4.7.11(1)
1,2	7	$ac + ad' = ad + ac'$	$= E(4, 5, 6)$
1,2	8	$b(c + d') = b(d + c')$	$= E(3)$
1	9	$b(c + d') = bc + bd'$	Th. 12.4.7.11(1)

1	10	$b(d + c') = bd + bc'$	Th. 12.4.7.11(1)
1,2	11	$bc + bd' = bd + bc'$	$= E(8, 9, 10)$
1	12	$(ac + bd) + (ad' + bc') = (ac + ad') + (bd + bc')$	Th. 12.4.4.10(1)
1,2	13	$bd + bc' = bc + bd'$	Th. 2.9.1.1(11)
1,2	14	$(ac + ad') + (bd + bc') = (ad + ac') + (bc + bd')$	$= E(7, 13)$
1	15	$(ad + ac') + (bc + bd') = (ad + bc) + (ac' + bd')$	Th. 12.4.4.10(1)
1,2	16	$(ac + bd) + (ad' + bc') = (ad + bc) + (ac' + bd')$	$= E(12, 14, 15)$
1,2	17	$[ac + bd, ad + bc]_{\mathbb{Z}} = [ac' + bd', ad' + bc']_{\mathbb{Z}}$	Th. 13.2.2.2(1,16)

Zusammenführung:

1	1	$a, b, a', b', c, d, c', d' \in \mathbb{N}$	A
2	2	$[a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}$	A
3	3	$[c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}}$	A
1,2	4	$[ac + bd, ad + bc]_{\mathbb{Z}} = [a'c + b'd, a'd + b'c]_{\mathbb{Z}}$	Th. 13.2.5.1(i)(1,2)
1,3	5	$[a'c + b'd, a'd + b'c]_{\mathbb{Z}} = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	Th. 13.2.5.1(ii)(1,3)
1,2,3	6	$[ac + bd, ad + bc]_{\mathbb{Z}} = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	Th. 2.9.1.2(4,5)

□

Theorem 13.2.5.2 (Quotientenabstieg der Multiplikation).

Mengen: $z, w, u, v, a, b, c, d, a', b', c', d', p, q, r, s$

$$z, w \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\ (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$$

Beweis.

Abgeschlossenheit der Ergebniskoordinaten Th. 13.2.5.2(i)

$$a, b, c, d \in \mathbb{N} \vdash ac + bd \in \mathbb{N} \wedge ad + bc \in \mathbb{N}$$

1	1	$a, b, c, d \in \mathbb{N}$	A
1	2	$ac \in \mathbb{N}$	Th. 12.4.7.3(1)
1	3	$bd \in \mathbb{N}$	Th. 12.4.7.3(1)
1	4	$ad \in \mathbb{N}$	Th. 12.4.7.3(1)
1	5	$bc \in \mathbb{N}$	Th. 12.4.7.3(1)
1	6	$ac + bd \in \mathbb{N}$	Th. 12.4.4.1(2,3)
1	7	$ad + bc \in \mathbb{N}$	Th. 12.4.4.1(4,5)
1	8	$ac + bd \in \mathbb{N} \wedge ad + bc \in \mathbb{N}$	$\wedge I(6, 7)$

Ergebniskandidat fester Repräsentanten Th. 13.2.5.2(ii)

$$\begin{aligned}
& a, b, c, d \in \mathbb{N}, \quad z = [a, b]_{\mathbb{Z}}, \quad w = [c, d]_{\mathbb{Z}} \vdash \\
& [ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\
& (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [ac + bd, ad + bc]_{\mathbb{Z}} = [pr + qs, ps + qr]_{\mathbb{Z}})
\end{aligned}$$

$$\begin{aligned}
& 1 \quad a, b, c, d \in \mathbb{N} \\
& 2 \quad z = [a, b]_{\mathbb{Z}} \\
& 3 \quad w = [c, d]_{\mathbb{Z}} \\
& 1 \quad 4 \quad ac + bd \in \mathbb{N} \wedge ad + bc \in \mathbb{N} \\
& 1 \quad 5 \quad [ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z} \\
& \quad 6 \quad [ac + bd, ad + bc]_{\mathbb{Z}} = [ac + bd, ad + bc]_{\mathbb{Z}} \\
& 1,2,3 \quad 7 \quad \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [ac + bd, ad + bc]_{\mathbb{Z}} = [pr + qs, ps + qr]_{\mathbb{Z}}) \\
& 1,2,3 \quad 8 \quad [ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [ac + bd, ad + bc]_{\mathbb{Z}} = [pr + qs, ps + qr]_{\mathbb{Z}})
\end{aligned}$$

Existenz durch getrennte Repräsentantenwahl

Th. 13.2.5.2(iii)

$$\begin{aligned}
& z, w \in \mathbb{Z} \vdash \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} \\
& (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})
\end{aligned}$$

$$\begin{aligned}
& 1 \quad 1 \quad z, w \in \mathbb{Z} \\
& 1 \quad 2 \quad z \in \mathbb{Z} \\
& 1 \quad 3 \quad w \in \mathbb{Z} \\
& 1 \quad 4 \quad \exists a \in \mathbb{N} \exists b \in \mathbb{N} z = [a, b]_{\mathbb{Z}} \\
& 1 \quad 5 \quad \exists c \in \mathbb{N} \exists d \in \mathbb{N} w = [c, d]_{\mathbb{Z}} \\
& 6 \quad 6 \quad a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \\
& 7 \quad 7 \quad c, d \in \mathbb{N} \wedge w = [c, d]_{\mathbb{Z}} \\
& 6 \quad 8 \quad a, b \in \mathbb{N} \\
& 6 \quad 9 \quad z = [a, b]_{\mathbb{Z}} \\
& 7 \quad 10 \quad c, d \in \mathbb{N} \\
& 7 \quad 11 \quad w = [c, d]_{\mathbb{Z}} \\
& 6,7 \quad 12 \quad a, b, c, d \in \mathbb{N} \\
& 6,7 \quad 13 \quad [ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge [ac + bd, ad + bc]_{\mathbb{Z}} = [pr + qs, ps + qr]_{\mathbb{Z}}) \\
& 6,7 \quad 14 \quad \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}}) \\
& \quad 6 \quad 15 \quad \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}}) \\
& 1 \quad 16 \quad \exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})
\end{aligned}$$

Eindeutigkeit für feste Repräsentanten

Th. 13.2.5.2(iv)

$$\begin{aligned}
& a, b, c, d, a', b', c', d' \in \mathbb{N}, \\
& z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}}, \\
& z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}} \vdash u = v
\end{aligned}$$

1	1	$a, b, c, d, a', b', c', d' \in \mathbb{N}$	A
2	2	$z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}}$	A
3	3	$z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	A
2	4	$z = [a, b]_{\mathbb{Z}}$	$\wedge E1(2)$
2	5	$w = [c, d]_{\mathbb{Z}}$	$\wedge E1(\wedge E2(2))$
2	6	$u = [ac + bd, ad + bc]_{\mathbb{Z}}$	$\wedge E2(\wedge E2(2))$
3	7	$z = [a', b']_{\mathbb{Z}}$	$\wedge E1(3)$
3	8	$w = [c', d']_{\mathbb{Z}}$	$\wedge E1(\wedge E2(3))$
3	9	$v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	$\wedge E2(\wedge E2(3))$
2,3	10	$[a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}$	$= E(4, 7)$
2,3	11	$[c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}}$	$= E(5, 8)$
1,2,3	12	$[ac + bd, ad + bc]_{\mathbb{Z}} = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	Th. 13.2.5.1(1,10,11)
1,2,3	13	$u = v$	$= E(6, 9, 12)$

Eindeutigkeit beliebiger Ergebniszeugen

Th. 13.2.5.2(v)

$(u \in \mathbb{Z} \wedge \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}})),$
 $(v \in \mathbb{Z} \wedge \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}})) \vdash u = v$

1	1	$u \in \mathbb{Z} \wedge \exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}})$	A
2	2	$v \in \mathbb{Z} \wedge \exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}})$	A
1	3	$\exists a, b, c, d \in \mathbb{N} (z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}})$	$\wedge E2(1)$
2	4	$\exists a', b', c', d' \in \mathbb{N} (z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}})$	$\wedge E2(2)$
5	5	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}}$	A
6	6	$a', b', c', d' \in \mathbb{N} \wedge z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	A
5	7	$a, b, c, d \in \mathbb{N}$	$\wedge E1(5)$
5	8	$z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [ac + bd, ad + bc]_{\mathbb{Z}}$	$\wedge E2(5)$
6	9	$a', b', c', d' \in \mathbb{N}$	$\wedge E1(6)$
6	10	$z = [a', b']_{\mathbb{Z}} \wedge w = [c', d']_{\mathbb{Z}} \wedge v = [a'c' + b'd', a'd' + b'c']_{\mathbb{Z}}$	$\wedge E2(6)$
5,6	11	$a, b, c, d, a', b', c', d' \in \mathbb{N}$	$\wedge I(7, 9)$
5,6	12	$u = v$	Th. 13.2.5.2(iv)(11,8,1)
5	13	$u = v$	$\exists E(4, 6, 12)$
1,2	14	$u = v$	$\exists E(3, 5, 13)$

Abschluss des Abstiegs

Th. 13.2.5.2(vi)

$z, w \in \mathbb{Z} \vdash \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N}$
 $(z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$

1	1	$z, w \in \mathbb{Z}$	A
1	2	$\exists u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}})$	Th. 13.2.5.2(iii)(1)

$$\begin{array}{l}
3 \quad 3 \quad u \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}}) A \\
4 \quad 4 \quad v \in \mathbb{Z} \wedge \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge v = [pr + qs, ps + qr]_{\mathbb{Z}}) A \\
3,4 \quad 5 \quad u = v \quad \text{Th. 13.2.5.2(v)(3,4)} \\
1 \quad 6 \quad \exists! u \in \mathbb{Z} \exists p \in \mathbb{N} \exists q \in \mathbb{N} \exists r \in \mathbb{N} \exists s \in \mathbb{N} (z = [p, q]_{\mathbb{Z}} \wedge w = [r, s]_{\mathbb{Z}} \wedge u = [pr + qs, ps + qr]_{\mathbb{Z}}) \exists! I(2, 3, 4, 5)
\end{array}$$

□

Der Quotientenabstieg liefert damit auch für die Multiplikation unabhängig von beiden gewählten Differenzenpaaren genau ein Ergebnis in $\mathbb{Z}$.

Definition 13.2.5.1 (Multiplikation ganzer Zahlen).

Mengen: a, b, c, d
Neue Symbole: $\cdot$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} := [a \cdot c + b \cdot d, a \cdot d + b \cdot c]_{\mathbb{Z}}$$

Theorem 13.2.5.3 (Abgeschlossenheit der Multiplikation ganzer Zahlen).

Mengen: z, w, a, b, c, d

$$z \in \mathbb{Z}, w \in \mathbb{Z} \vdash z \cdot w \in \mathbb{Z}$$

Beweis.

Abgeschlossenheit für feste Repräsentanten Th. 13.2.5.3(i)

$$\begin{array}{l}
a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \\
\vdash z \cdot w \in \mathbb{Z}
\end{array}$$

$$\begin{array}{ll}
1 \quad 1 \quad a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} & A \\
1 \quad 2 \quad z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}} & \text{Def. 13.2.5.1(1)} \\
1 \quad 3 \quad ac + bd \in \mathbb{N} & \text{Th. 12.4.4.1(Th. 12.4.7.3(1), Th. 12.4.7.3(1))} \\
1 \quad 4 \quad ad + bc \in \mathbb{N} & \text{Th. 12.4.4.1(Th. 12.4.7.3(1), Th. 12.4.7.3(1))} \\
1 \quad 5 \quad [ac + bd, ad + bc]_{\mathbb{Z}} \in \mathbb{Z} & \text{Th. 13.2.2.1(3,4)} \\
1 \quad 6 \quad z \cdot w \in \mathbb{Z} & = E(2, 5)
\end{array}$$

Elimination der Repräsentanten Th. 13.2.5.3(ii)

$$z \in \mathbb{Z}, w \in \mathbb{Z} \vdash z \cdot w \in \mathbb{Z}$$

$$\begin{array}{ll}
1 \quad 1 \quad z \in \mathbb{Z} & A \\
2 \quad 2 \quad w \in \mathbb{Z} & A
\end{array}$$

1	3	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(1)
2	4	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.3(2)
5	5	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A$	
5	6	$z \cdot w \in \mathbb{Z}$	Th. 13.2.5.3(i)(5)
1,2	7	$z \cdot w \in \mathbb{Z}$	$\exists E(3, 4, 5, 6)$

□

Theorem 13.2.5.4 (Kommutativität der Multiplikation ganzer Zahlen).

Mengen: z, w, a, b, c, d

$$z, w \in \mathbb{Z} \vdash z \cdot w = w \cdot z$$

Beweis.

Produkte gewählter Repräsentanten Th. 13.2.5.4(i)

$$\begin{aligned} a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \\ \vdash z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}} \wedge w \cdot z = [ca + db, cb + da]_{\mathbb{Z}} \end{aligned}$$

1	1	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}}$	A
1	2	$z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}}$	Def. 13.2.5.1(1)
1	3	$w \cdot z = [ca + db, cb + da]_{\mathbb{Z}}$	Def. 13.2.5.1(1)
1	4	$z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}} \wedge w \cdot z = [ca + db, cb + da]_{\mathbb{Z}} \wedge I(2, 3)$	

Vertauschung der Klassenkoordinaten Th. 13.2.5.4(ii)

$$a, b, c, d \in \mathbb{N} \vdash [ac + bd, ad + bc]_{\mathbb{Z}} = [ca + db, cb + da]_{\mathbb{Z}}$$

1	1	$a, b, c, d \in \mathbb{N}$	A
1	2	$ac = ca$	Th. 12.4.7.9(1)
1	3	$bd = db$	Th. 12.4.7.9(1)
1	4	$ad = da$	Th. 12.4.7.9(1)
1	5	$bc = cb$	Th. 12.4.7.9(1)
1	6	$ac + bd = ca + db$	$= E(2, 3)$
1	7	$ad + bc = cb + da$	Th. 12.4.4.8(= $E(4, 5)$)
1	8	$[ac + bd, ad + bc]_{\mathbb{Z}} = [ca + db, cb + da]_{\mathbb{Z}} = E(6, 7)$	

Elimination der Repräsentanten Th. 13.2.5.4(iii)

$$z, w \in \mathbb{Z} \vdash z \cdot w = w \cdot z$$

1	$1 \ z, w \in \mathbb{Z}$	A
1	$2 \ z \in \mathbb{Z}$	$\wedge E1(1)$
1	$3 \ w \in \mathbb{Z}$	$\wedge E2(1)$
1	$4 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(2)
1	$5 \ \exists c, d \in \mathbb{N} \ w = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.3(3)
6	$6 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}}$	A
6	$7 \ z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}} \wedge w \cdot z = [ca + db, cb + da]_{\mathbb{Z}}$	Th. 13.2.5.4(i)(6)
6	$8 \ z \cdot w = [ac + bd, ad + bc]_{\mathbb{Z}}$	$\wedge E1(7)$
6	$9 \ w \cdot z = [ca + db, cb + da]_{\mathbb{Z}}$	$\wedge E2(7)$
6	$10 \ [ac + bd, ad + bc]_{\mathbb{Z}} = [ca + db, cb + da]_{\mathbb{Z}}$	Th. 13.2.5.4(ii)(6)
6	$11 \ z \cdot w = w \cdot z$	$= E(8, 9, 10)$
1	$12 \ z \cdot w = w \cdot z$	$\exists E(4, 5, 6, 11)$

□

Theorem 13.2.5.5 (Einheitsgesetze der Multiplikation ganzer Zahlen).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash z \cdot 1_{\mathbb{Z}} = z \wedge 1_{\mathbb{Z}} \cdot z = z$$

Beweis.

Rechte Einheit am Repräsentanten

Th. 13.2.5.5(i)

$$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \vdash z \cdot 1_{\mathbb{Z}} = z$$

1	$1 \ a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
	$2 \ 1_{\mathbb{Z}} = \iota_{\mathbb{Z}}(1)$	Def. 13.2.3.3
	$3 \ \iota_{\mathbb{Z}}(1) = [1, 0]_{\mathbb{Z}}$	Def. 13.2.3.1(Th. 12.4.4.11)
1	$4 \ z \cdot 1_{\mathbb{Z}} = [a \cdot 1 + b \cdot 0, a \cdot 0 + b \cdot 1]_{\mathbb{Z}}$	Def. 13.2.5.1(1,2,3)
1	$5 \ a \cdot 1 = a$	Th. 12.4.7.7(1)
1	$6 \ b \cdot 1 = b$	Th. 12.4.7.7(1)
1	$7 \ a \cdot 0 = 0$	Th. 12.4.7.4(1)
1	$8 \ b \cdot 0 = 0$	Th. 12.4.7.4(1)
1	$9 \ a \cdot 1 + b \cdot 0 = a$	$= E(5, 8, Th. 12.4.4.2(1))$
1	$10 \ a \cdot 0 + b \cdot 1 = b$	$= E(7, 6, Th. 12.4.4.4(1))$
1	$11 \ z \cdot 1_{\mathbb{Z}} = z$	$= E(1, 4, 9, 10)$

Beidseitige Einheit

Th. 13.2.5.5(ii)

$$z \in \mathbb{Z} \vdash z \cdot 1_{\mathbb{Z}} = z \wedge 1_{\mathbb{Z}} \cdot z = z$$

- 1 1 $z \in \mathbb{Z}$ A
- 1 2 $\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$ Th. 13.2.2.3(1)
- 3 3 $a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$ A
- 3 4 $z \cdot 1_{\mathbb{Z}} = z$ Th. 13.2.5.5(i)(3)
- 3 5 $1_{\mathbb{Z}} \cdot z = z$ Th. 13.2.5.4(Th. 13.2.3.5,1, Th. 2.9.1.1(4))
- 3 6 $z \cdot 1_{\mathbb{Z}} = z \wedge 1_{\mathbb{Z}} \cdot z = z \wedge I(4, 5)$
- 1 7 $z \cdot 1_{\mathbb{Z}} = z \wedge 1_{\mathbb{Z}} \cdot z = z \exists E(2, 3, 6)$

□

Theorem 13.2.5.6 (Assoziativität der Multiplikation ganzer Zahlen).

Mengen: $z, w, u, a, b, c, d, e, f$

$$z, w, u \in \mathbb{Z} \vdash (z \cdot w) \cdot u = z \cdot (w \cdot u)$$

Beweis.

Positive Koordinate: linke Seite ausmultiplizieren Th. 13.2.5.6(i)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$(ac + bd)e + (ad + bc)f = ((ac)e + (bd)e) + ((ad)f + (bc)f)$$

- 1 1 $a, b, c, d, e, f \in \mathbb{N}$ A
- 1 2 $(ac + bd)e = (ac)e + (bd)e$ Th. 12.4.7.12(1)
- 1 3 $(ad + bc)f = (ad)f + (bc)f$ Th. 12.4.7.12(1)
- 1 4 $(ac + bd)e + (ad + bc)f = ((ac)e + (bd)e) + ((ad)f + (bc)f) = E(2, 3)$

Positive Koordinate: rechte Seite ausmultiplizieren Th. 13.2.5.6(ii)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$a(ce + df) + b(cf + de) = (a(ce) + a(df)) + (b(cf) + b(de))$$

- 1 1 $a, b, c, d, e, f \in \mathbb{N}$ A
- 1 2 $a(ce + df) = a(ce) + a(df)$ Th. 12.4.7.11(1)
- 1 3 $b(cf + de) = b(cf) + b(de)$ Th. 12.4.7.11(1)
- 1 4 $a(ce + df) + b(cf + de) = (a(ce) + a(df)) + (b(cf) + b(de)) = E(2, 3)$

Positive Koordinate: ausmultiplizierte Terme ordnen Th. 13.2.5.6(iii)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$((ac)e + (bd)e) + ((ad)f + (bc)f)$$

$$= (a(ce) + a(df)) + (b(cf) + b(de))$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac)e = a(ce)$	Th. 12.4.7.13(1)
1	3	$(bd)e = b(de)$	Th. 12.4.7.13(1)
1	4	$(ad)f = a(df)$	Th. 12.4.7.13(1)
1	5	$(bc)f = b(cf)$	Th. 12.4.7.13(1)
1	6	$((ac)e + (bd)e) + ((ad)f + (bc)f) = (a(ce) + b(de)) + (a(df) + b(cf)) = E(2, 3, 4, 5)$	
1	7	$(a(ce) + b(de)) + (a(df) + b(cf)) = (a(ce) + a(df)) + (b(de) + b(cf))$	Th. 12.4.4.10(1)
1	8	$b(de) + b(cf) = b(cf) + b(de)$	Th. 12.4.4.8(1)
1	9	$(a(ce) + a(df)) + (b(de) + b(cf)) = (a(ce) + a(df)) + (b(cf) + b(de)) = E(8)$	
1	10	$((ac)e + (bd)e) + ((ad)f + (bc)f) = (a(ce) + a(df)) + (b(cf) + b(de)) = E(6, 7, 9)$	

Gleichheit der positiven Koordinaten

Th. 13.2.5.6(iv)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$(ac + bd)e + (ad + bc)f = a(ce + df) + b(cf + de)$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac + bd)e + (ad + bc)f = ((ac)e + (bd)e) + ((ad)f + (bc)f)$	Th. 13.2.5.6(i)(1)
1	3	$((ac)e + (bd)e) + ((ad)f + (bc)f) = (a(ce) + a(df)) + (b(cf) + b(de))$	Th. 13.2.5.6(iii)(1)
1	4	$a(ce + df) + b(cf + de) = (a(ce) + a(df)) + (b(cf) + b(de))$	Th. 13.2.5.6(ii)(1)
1	5	$(a(ce) + a(df)) + (b(cf) + b(de)) = a(ce + df) + b(cf + de)$	Th. 2.9.1.1(4)
1	6	$(ac + bd)e + (ad + bc)f = a(ce + df) + b(cf + de)$	$= E(2, 3, 5)$

Negative Koordinate: linke Seite ausmultiplizieren

Th. 13.2.5.6(v)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$(ac + bd)f + (ad + bc)e = ((ac)f + (bd)f) + ((ad)e + (bc)e)$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac + bd)f = (ac)f + (bd)f$	Th. 12.4.7.12(1)
1	3	$(ad + bc)e = (ad)e + (bc)e$	Th. 12.4.7.12(1)
1	4	$(ac + bd)f + (ad + bc)e = ((ac)f + (bd)f) + ((ad)e + (bc)e) = E(2, 3)$	

Negative Koordinate: rechte Seite ausmultiplizieren

Th. 13.2.5.6(vi)

$$a, b, c, d, e, f \in \mathbb{N} \vdash$$

$$a(cf + de) + b(ce + df) = (a(cf) + a(de)) + (b(ce) + b(df))$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$a(cf + de) = a(cf) + a(de)$	Th. 12.4.7.11(1)
1	3	$b(ce + df) = b(ce) + b(df)$	Th. 12.4.7.11(1)
1	4	$a(cf + de) + b(ce + df) = (a(cf) + a(de)) + (b(ce) + b(df)) = E(2, 3)$	

Negative Koordinate: ausmultiplizierte Terme ordnen

Th. 13.2.5.6(vii)

$$\begin{aligned}
& a, b, c, d, e, f \in \mathbb{N} \vdash \\
& ((ac)f + (bd)f) + ((ad)e + (bc)e) \\
& = (a(cf) + a(de)) + (b(ce) + b(df))
\end{aligned}$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac)f = a(cf)$	Th. 12.4.7.13(1)
1	3	$(bd)f = b(df)$	Th. 12.4.7.13(1)
1	4	$(ad)e = a(de)$	Th. 12.4.7.13(1)
1	5	$(bc)e = b(ce)$	Th. 12.4.7.13(1)
1	6	$((ac)f + (bd)f) + ((ad)e + (bc)e) = (a(cf) + b(df)) + (a(de) + b(ce)) = E(2, 3, 4, 5)$	
1	7	$(a(cf) + b(df)) + (a(de) + b(ce)) = (a(cf) + a(de)) + (b(df) + b(ce))$	Th. 12.4.4.10(1)
1	8	$b(df) + b(ce) = b(ce) + b(df)$	Th. 12.4.4.8(1)
1	9	$(a(cf) + a(de)) + (b(df) + b(ce)) = (a(cf) + a(de)) + (b(ce) + b(df)) = E(8)$	
1	10	$((ac)f + (bd)f) + ((ad)e + (bc)e) = (a(cf) + a(de)) + (b(ce) + b(df)) = E(6, 7, 9)$	

Gleichheit der negativen Koordinaten

Th. 13.2.5.6(viii)

$$\begin{aligned}
& a, b, c, d, e, f \in \mathbb{N} \vdash \\
& (ac + bd)f + (ad + bc)e = a(cf + de) + b(ce + df)
\end{aligned}$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$(ac + bd)f + (ad + bc)e = ((ac)f + (bd)f) + ((ad)e + (bc)e)$	Th. 13.2.5.6(v)(1)
1	3	$((ac)f + (bd)f) + ((ad)e + (bc)e) = (a(cf) + a(de)) + (b(ce) + b(df))$	Th. 13.2.5.6(vii)(1)
1	4	$a(cf + de) + b(ce + df) = (a(cf) + a(de)) + (b(ce) + b(df))$	Th. 13.2.5.6(vi)(1)
1	5	$(a(cf) + a(de)) + (b(ce) + b(df)) = a(cf + de) + b(ce + df)$	Th. 2.9.1.1(4)
1	6	$(ac + bd)f + (ad + bc)e = a(cf + de) + b(ce + df)$	$= E(2, 3, 5)$

Rechnung mit Klassen

Th. 13.2.5.6(ix)

$$\begin{aligned}
& a, b, c, d, e, f \in \mathbb{N} \vdash \\
& ([a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}}) \cdot [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}})
\end{aligned}$$

1	1	$a, b, c, d, e, f \in \mathbb{N}$	A
1	2	$([a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}}) \cdot [e, f]_{\mathbb{Z}} = [(ac + bd)e + (ad + bc)f, (ac + bd)f + (ad + bc)e]_{\mathbb{Z}}$	Def. 13.2.5.1(1)
1	3	$[a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}}) = [a(ce + df) + b(cf + de), a(cf + de) + b(ce + df)]_{\mathbb{Z}}$	Def. 13.2.5.1(1)
1	4	$(ac + bd)e + (ad + bc)f = a(ce + df) + b(cf + de)$	Th. 13.2.5.6(iv)(1)
1	5	$(ac + bd)f + (ad + bc)e = a(cf + de) + b(ce + df)$	Th. 13.2.5.6(viii)(1)
1	6	$([a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}}) \cdot [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}})$	$= E(2, 3, 4, 5)$

Schluss:

1	$1 \ z, w, u \in \mathbb{Z}$	A
1	$2 \ z \in \mathbb{Z}$	$\wedge E1(1)$
1	$3 \ w \in \mathbb{Z}$	$\wedge E1(\wedge E2(1))$
1	$4 \ u \in \mathbb{Z}$	$\wedge E2(\wedge E2(1))$
1	$5 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(2)
1	$6 \ \exists c, d \in \mathbb{N} \ w = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.3(3)
1	$7 \ \exists e, f \in \mathbb{N} \ u = [e, f]_{\mathbb{Z}}$	Th. 13.2.2.3(4)
8	$8 \ a, b, c, d, e, f \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [e, f]_{\mathbb{Z}} A$	
8	$9 \ ([a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}}) \cdot [e, f]_{\mathbb{Z}} = [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}})$	Th. 13.2.5.6(ix)(8)
8	$10 \ (z \cdot w) \cdot u = z \cdot (w \cdot u)$	$= E(8, 9)$
1	$11 \ (z \cdot w) \cdot u = z \cdot (w \cdot u)$	$\exists E(5, 6, 7, 8, 10)$

□

Theorem 13.2.5.7 (Links distributivität ganzer Zahlen).

Mengen: $z, w, u, a, b, c, d, e, f$

$$z, w, u \in \mathbb{Z} \vdash z \cdot (w + u) = z \cdot w + z \cdot u$$

Beweis.

Positive Koordinate distributiv ausmultiplizieren Th. 13.2.5.7(i)

$$\begin{aligned} a, b, c, d, e, f \in \mathbb{N} \vdash \\ a(c + e) + b(d + f) &= (ac + bd) + (ae + bf) \end{aligned}$$

1	$1 \ a, b, c, d, e, f \in \mathbb{N}$	A
1	$2 \ a(c + e) = ac + ae$	Th. 12.4.7.11(1)
1	$3 \ b(d + f) = bd + bf$	Th. 12.4.7.11(1)
1	$4 \ a(c + e) + b(d + f) = (ac + ae) + (bd + bf)$	$= E(2, 3)$
1	$5 \ (ac + ae) + (bd + bf) = (ac + bd) + (ae + bf)$	Th. 12.4.4.10(1)
1	$6 \ a(c + e) + b(d + f) = (ac + bd) + (ae + bf)$	$= E(4, 5)$

Negative Koordinate distributiv ausmultiplizieren Th. 13.2.5.7(ii)

$$\begin{aligned} a, b, c, d, e, f \in \mathbb{N} \vdash \\ a(d + f) + b(c + e) &= (ad + bc) + (af + be) \end{aligned}$$

1	$1 \ a, b, c, d, e, f \in \mathbb{N}$	A
1	$2 \ a(d + f) = ad + af$	Th. 12.4.7.11(1)
1	$3 \ b(c + e) = bc + be$	Th. 12.4.7.11(1)

$$\begin{array}{l}
1 \quad 4 \quad a(d+f) + b(c+e) = (ad+af) + (bc+be) = E(2,3) \\
1 \quad 5 \quad (ad+af) + (bc+be) = (ad+bc) + (af+be) \text{Th. 12.4.4.10(1)} \\
1 \quad 6 \quad a(d+f) + b(c+e) = (ad+bc) + (af+be) = E(4,5)
\end{array}$$

Distributivrechnung auf Klassen

Th. 13.2.5.7(iii)

$$\begin{array}{l}
a, b, c, d, e, f \in \mathbb{N} \vdash \\
[a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} + [a, b]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}}
\end{array}$$

$$\begin{array}{ll}
1 \quad 1 \quad a, b, c, d, e, f \in \mathbb{N} & A \\
1 \quad 2 \quad [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a(c+e) + b(d+f), a(d+f) + b(c+e)]_{\mathbb{Z}} & \text{Def. 13.2.4.2(Def. 13.2.5.1(1))} \\
1 \quad 3 \quad [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} + [a, b]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}} = [(ac+bd) + (ae+bf), (ad+bc) + (af+be)]_{\mathbb{Z}} & \text{Def. 13.2.5.1(Def. 13.2.4.2(1))} \\
1 \quad 4 \quad a(c+e) + b(d+f) = (ac+bd) + (ae+bf) & \text{Th. 13.2.5.7(i)(1)} \\
1 \quad 5 \quad a(d+f) + b(c+e) = (ad+bc) + (af+be) & \text{Th. 13.2.5.7(ii)(1)} \\
1 \quad 6 \quad [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} + [a, b]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}} & = E(2, 3, 4, 5)
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 \quad 1 \quad z, w, u \in \mathbb{Z} & A \\
1 \quad 2 \quad z \in \mathbb{Z} & \wedge E1(1) \\
1 \quad 3 \quad w \in \mathbb{Z} & \wedge E1(\wedge E2(1)) \\
1 \quad 4 \quad u \in \mathbb{Z} & \wedge E2(\wedge E2(1)) \\
1 \quad 5 \quad \exists a, b \in \mathbb{N} \quad z = [a, b]_{\mathbb{Z}} & \text{Th. 13.2.2.3(2)} \\
1 \quad 6 \quad \exists c, d \in \mathbb{N} \quad w = [c, d]_{\mathbb{Z}} & \text{Th. 13.2.2.3(3)} \\
1 \quad 7 \quad \exists e, f \in \mathbb{N} \quad u = [e, f]_{\mathbb{Z}} & \text{Th. 13.2.2.3(4)} \\
8 \quad 8 \quad a, b, c, d, e, f \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [e, f]_{\mathbb{Z}} & A \\
8 \quad 9 \quad [a, b]_{\mathbb{Z}} \cdot ([c, d]_{\mathbb{Z}} + [e, f]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} \cdot [c, d]_{\mathbb{Z}} + [a, b]_{\mathbb{Z}} \cdot [e, f]_{\mathbb{Z}} & \text{Th. 13.2.5.7(iii)(8)} \\
8 \quad 10 \quad z \cdot (w + u) = z \cdot w + z \cdot u & = E(8, 9) \\
1 \quad 11 \quad z \cdot (w + u) = z \cdot w + z \cdot u & \exists E(5, 6, 7, 8, 10)
\end{array}$$

□

Theorem 13.2.5.8 (Rechtsdistributivität ganzer Zahlen).

Mengen: z, w, u

$$z, w, u \in \mathbb{Z} \vdash (z + w) \cdot u = z \cdot u + w \cdot u$$

$$\begin{array}{ll}
1 \quad 1 \quad z, w, u \in \mathbb{Z} & A \\
1 \quad 2 \quad (z + w) \cdot u = u \cdot (z + w) & \text{Th. 13.2.5.4(Th. 13.2.4.7(1), 1)} \\
1 \quad 3 \quad u \cdot (z + w) = u \cdot z + u \cdot w & \text{Th. 13.2.5.7(1)} \\
1 \quad 4 \quad u \cdot z = z \cdot u & \text{Th. 13.2.5.4(1)}
\end{array}$$

$$\begin{array}{ll}
1 \ 5 \ u \cdot w = w \cdot u & Th. \ 13.2.5.4(1) \\
1 \ 6 \ u \cdot z + u \cdot w = z \cdot u + w \cdot u = E(4, 5) \\
1 \ 7 \ (z + w) \cdot u = z \cdot u + w \cdot u = E(2, 3, 6)
\end{array}$$

2.6 Verträglichkeit der Einbettung mit der Arithmetik

Theorem 13.2.6.1 (Additivität der natürlichen Einbettung).

Mengen: m, n

Funktion: $\iota_{\mathbb{Z}}$

$$m, n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(m + n) = \iota_{\mathbb{Z}}(m) + \iota_{\mathbb{Z}}(n)$$

$$\begin{array}{ll}
1 \ 1 \ m, n \in \mathbb{N} & A \\
1 \ 2 \ m + n \in \mathbb{N} & Th. \ 12.4.4.1(1) \\
1 \ 3 \ \iota_{\mathbb{Z}}(m + n) = [m + n, 0]_{\mathbb{Z}} & Def. \ 13.2.3.1(2) \\
1 \ 4 \ \iota_{\mathbb{Z}}(m) = [m, 0]_{\mathbb{Z}} & Def. \ 13.2.3.1(1) \\
1 \ 5 \ \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}} & Def. \ 13.2.3.1(1) \\
1 \ 6 \ \iota_{\mathbb{Z}}(m) + \iota_{\mathbb{Z}}(n) = [m + n, 0 + 0]_{\mathbb{Z}} & Def. \ 13.2.4.2(1, 4, 5) \\
7 \ 0 + 0 = 0 & Th. \ 12.4.4.2(Ax. \ 12.3.1) \\
1 \ 8 \ \iota_{\mathbb{Z}}(m) + \iota_{\mathbb{Z}}(n) = [m + n, 0]_{\mathbb{Z}} & = E(7, 6) \\
1 \ 9 \ \iota_{\mathbb{Z}}(m + n) = \iota_{\mathbb{Z}}(m) + \iota_{\mathbb{Z}}(n) & Th. \ 2.9.1.3(3, 8)
\end{array}$$

Theorem 13.2.6.2 (Multiplikativität der natürlichen Einbettung).

Mengen: m, n

Funktion: $\iota_{\mathbb{Z}}$

$$m, n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(m \cdot n) = \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n)$$

Beweis.

Produkt der eingebetteten Faktoren

Th. 13.2.6.2(i)

$$m, n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) = [mn, 0]_{\mathbb{Z}}$$

$$\begin{array}{ll}
1 \ 1 \ m, n \in \mathbb{N} & A \\
1 \ 2 \ mn \in \mathbb{N} & Th. \ 12.4.7.3(1) \\
1 \ 3 \ \iota_{\mathbb{Z}}(m) = [m, 0]_{\mathbb{Z}} & Def. \ 13.2.3.1(1) \\
1 \ 4 \ \iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}} & Def. \ 13.2.3.1(1) \\
1 \ 5 \ \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) = [mn + 0 \cdot 0, m \cdot 0 + 0 \cdot n]_{\mathbb{Z}} & Def. \ 13.2.5.1(1, 3, 4)
\end{array}$$

	$6 \ 0 \cdot 0 = 0$	Th. 12.4.7.6(Ax. 12.3.1)
1	$7 \ m \cdot 0 = 0$	Th. 12.4.7.4(1)
1	$8 \ 0 \cdot n = 0$	Th. 12.4.7.6(1)
1	$9 \ mn + 0 \cdot 0 = mn$	$= E(6, Th. 12.4.4.2(2))$
1	$10 \ m \cdot 0 + 0 \cdot n = 0$	$= E(7, 8, Th. 12.4.4.2(Ax. 12.3.1))$
1	$11 \ \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) = [mn, 0]_{\mathbb{Z}}$	$= E(5, 9, 10)$

Vergleich mit dem Produktbild

Th. 13.2.6.2(ii)

$$m, n \in \mathbb{N} \vdash \iota_{\mathbb{Z}}(m \cdot n) = \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n)$$

1	$1 \ m, n \in \mathbb{N}$	A
1	$2 \ mn \in \mathbb{N}$	Th. 12.4.7.3(1)
1	$3 \ \iota_{\mathbb{Z}}(mn) = [mn, 0]_{\mathbb{Z}}$	Def. 13.2.3.1(2)
1	$4 \ \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n) = [mn, 0]_{\mathbb{Z}}$	Th. 13.2.6.2(i)(1)
1	$5 \ \iota_{\mathbb{Z}}(mn) = \iota_{\mathbb{Z}}(m) \cdot \iota_{\mathbb{Z}}(n)$	Th. 2.9.1.3(3,4)

□

Kapitel 3

Nachfolger, Vorgänger und Normalformen

3.1 Nachfolger und Vorgänger

Definition 13.3.1.1 (Nachfolger einer ganzen Zahl).

Mengen: $\mathbb{Z}$
Funktionensymbol: $S_{\mathbb{Z}}$
Neue Symbole: $S_{\mathbb{Z}}$

$$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}, \quad z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(z) := z + 1_{\mathbb{Z}}$$

$$\begin{array}{ll} 1 & 1 \quad z \in \mathbb{Z} \quad A \\ 2 & 1_{\mathbb{Z}} \in \mathbb{Z} \quad Th. \ 13.2.3.5 \\ 1 & 3 \quad z + 1_{\mathbb{Z}} \in \mathbb{Z} \quad Th. \ 13.2.4.7(1, 2) \end{array}$$

Definition 13.3.1.2 (Vorgänger einer ganzen Zahl).

Mengen: $\mathbb{Z}$
Funktionensymbol: $P_{\mathbb{Z}}$
Neue Symbole: $P_{\mathbb{Z}}$

$$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}, \quad z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(z) := z + (-1_{\mathbb{Z}})$$

$$\begin{array}{ll} 1 & 1 \quad z \in \mathbb{Z} \quad A \\ 2 & 1_{\mathbb{Z}} \in \mathbb{Z} \quad Th. \ 13.2.3.5 \\ 3 & -1_{\mathbb{Z}} \in \mathbb{Z} \quad Th. \ 13.2.4.3(2) \\ 1 & 4 \quad z + (-1_{\mathbb{Z}}) \in \mathbb{Z} \quad Th. \ 13.2.4.7(1, 3) \end{array}$$

Theorem 13.3.1.1 (Nachfolgerfunktion auf den ganzen Zahlen).

Funktion: $S_{\mathbb{Z}}$

$$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$1 \ S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \text{ Def. 13.3.1.1}$$

Theorem 13.3.1.2 (Vorgängerfunktion auf den ganzen Zahlen).

Funktion: $P_{\mathbb{Z}}$

$$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$1 \ P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \text{ Def. 13.3.1.2}$$

Theorem 13.3.1.3 (Nachfolger auf Klassen).

Mengen: a, b

$$a, b \in \mathbb{N} \vdash S_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [\text{succ}(a), b]_{\mathbb{Z}}$$

$$\begin{array}{ll} 1 \ 1 \ a, b \in \mathbb{N} & A \\ 1 \ 2 \ [a, b]_{\mathbb{Z}} \in \mathbb{Z} & Th. \ 13.2.2.1(1) \\ 1 \ 3 \ S_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} + 1_{\mathbb{Z}} & Def. \ 13.3.1.1(2) \\ \quad 4 \ 1_{\mathbb{Z}} = [1, 0]_{\mathbb{Z}} & Def. \ 13.2.3.3(Def. \ 13.2.3.1(Th. \ 12.4.4.11)) \\ 1 \ 5 \ [a, b]_{\mathbb{Z}} + 1_{\mathbb{Z}} = [a + 1, b + 0]_{\mathbb{Z}} & Def. \ 13.2.4.2(1, 4) \\ 1 \ 6 \ a + 1 = \text{succ}(a) & Th. \ 12.4.4.12(1) \\ 1 \ 7 \ b + 0 = b & Th. \ 12.4.4.2(1) \\ 1 \ 8 \ S_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [\text{succ}(a), b]_{\mathbb{Z}} & = E(3, 5, 6, 7) \end{array}$$

Theorem 13.3.1.4 (Vorgänger auf Klassen).

Mengen: a, b

$$a, b \in \mathbb{N} \vdash P_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [a, \text{succ}(b)]_{\mathbb{Z}}$$

1 1	$a, b \in \mathbb{N}$	A
1 2	$[a, b]_{\mathbb{Z}} \in \mathbb{Z}$	$Th. 13.2.2.1(1)$
1 3	$P_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [a, b]_{\mathbb{Z}} + (-1_{\mathbb{Z}})$	$Def. 13.3.1.2(2)$
4	$-1_{\mathbb{Z}} = [0, 1]_{\mathbb{Z}}$	$Def. 13.2.4.1(Def. 13.2.3.3, Def. 13.2.3.1)$
1 5	$[a, b]_{\mathbb{Z}} + (-1_{\mathbb{Z}}) = [a + 0, b + 1]_{\mathbb{Z}}$	$Def. 13.2.4.2(1, 4)$
1 6	$a + 0 = a$	$Th. 12.4.4.2(1)$
1 7	$b + 1 = \text{succ}(b)$	$Th. 12.4.4.12(1)$
1 8	$P_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) = [a, \text{succ}(b)]_{\mathbb{Z}}$	$= E(3, 5, 6, 7)$

Theorem 13.3.1.5 (Vorgänger nach Nachfolger).

Mengen: z

$$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$$

Beweis.

Auflösen der Definitionen

Th. 13.3.1.5(i)

$$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = (z + 1_{\mathbb{Z}}) + (-1_{\mathbb{Z}})$$

1 1	$z \in \mathbb{Z}$	A
1 2	$S_{\mathbb{Z}}(z) = z + 1_{\mathbb{Z}}$	Def. 13.3.1.1(1)
1 3	$S_{\mathbb{Z}}(z) \in \mathbb{Z}$	Th. 5.2.3.10(Th. 13.3.1.1, 1)
1 4	$P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = S_{\mathbb{Z}}(z) + (-1_{\mathbb{Z}})$	Def. 13.3.1.2(3)
1 5	$P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = (z + 1_{\mathbb{Z}}) + (-1_{\mathbb{Z}})$	$= E(2, 4)$

Schluss:

1 1	$z \in \mathbb{Z}$	A
1 2	$P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = (z + 1_{\mathbb{Z}}) + (-1_{\mathbb{Z}})$	Th. 13.3.1.5(i)(1)
3	$1_{\mathbb{Z}} \in \mathbb{Z}$	Th. 13.2.3.5
4	$1_{\mathbb{Z}} + (-1_{\mathbb{Z}}) = 0_{\mathbb{Z}}$	$\wedge E1(Th. 13.2.4.11(3))$
1 5	$(z + 1_{\mathbb{Z}}) + (-1_{\mathbb{Z}}) = z + (1_{\mathbb{Z}} + (-1_{\mathbb{Z}}))$	Th. 13.2.4.8(1, 3, Th. 13.2.4.3(3))
1 6	$z + (1_{\mathbb{Z}} + (-1_{\mathbb{Z}})) = z + 0_{\mathbb{Z}}$	$= E(4)$
1 7	$z + 0_{\mathbb{Z}} = z$	$\wedge E1(Th. 13.2.4.10(1))$
1 8	$P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$	$= E(2, 5, 6, 7)$

□

Theorem 13.3.1.6 (Nachfolger nach Vorgänger).

Mengen: z

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$$

Beweis.

Auflösen der Definitionen

Th. 13.3.1.6(i)

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = (z + (-1_{\mathbb{Z}})) + 1_{\mathbb{Z}}$$

1	$z \in \mathbb{Z}$	A
1	$2 \ P_{\mathbb{Z}}(z) = z + (-1_{\mathbb{Z}})$	Def. 13.3.1.2(1)
1	$3 \ P_{\mathbb{Z}}(z) \in \mathbb{Z}$	Th. 5.2.3.10(Th. 13.3.1.2,1)
1	$4 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = P_{\mathbb{Z}}(z) + 1_{\mathbb{Z}}$	Def. 13.3.1.1(3)
1	$5 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = (z + (-1_{\mathbb{Z}})) + 1_{\mathbb{Z}} = E(2, 4)$	

Assoziativität und Inverses

Th. 13.3.1.6(ii)

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$$

1	$z \in \mathbb{Z}$	A
1	$2 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = (z + (-1_{\mathbb{Z}})) + 1_{\mathbb{Z}}$	Th. 13.3.1.6(i)(1)
	$3 \ 1_{\mathbb{Z}} \in \mathbb{Z}$	Th. 13.2.3.5
	$4 \ -1_{\mathbb{Z}} \in \mathbb{Z}$	Th. 13.2.4.3(3)
	$5 \ (-1_{\mathbb{Z}}) + 1_{\mathbb{Z}} = 0_{\mathbb{Z}}$	$\wedge E2(Th. 13.2.4.11(3))$
1	$6 \ (z + (-1_{\mathbb{Z}})) + 1_{\mathbb{Z}} = z + ((-1_{\mathbb{Z}}) + 1_{\mathbb{Z}})$	Th. 13.2.4.8(1,4,3)
1	$7 \ z + ((-1_{\mathbb{Z}}) + 1_{\mathbb{Z}}) = z + 0_{\mathbb{Z}}$	$= E(5)$
1	$8 \ z + 0_{\mathbb{Z}} = z$	$\wedge E1(Th. 13.2.4.10(1))$
1	$9 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$	$= E(2, 6, 7, 8)$

□

Theorem 13.3.1.7 (Bijektivität des Ganzzahlnachfolgers).

Funktionen: $S_{\mathbb{Z}}, P_{\mathbb{Z}}$

$$S_{\mathbb{Z}}: \mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$$

Beweis.

Injektivität

Th. 13.3.1.7(i)

$$x, y \in \mathbb{Z}, S_{\mathbb{Z}}(x) = S_{\mathbb{Z}}(y) \vdash x = y$$

$$\begin{array}{ll}
1 & 1 \ x, y \in \mathbb{Z} \quad A \\
2 & 2 \ S_{\mathbb{Z}}(x) = S_{\mathbb{Z}}(y) \quad A \\
1 & 3 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(x)) = x \quad \text{Th. 13.3.1.5(1)} \\
1 & 4 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(y)) = y \quad \text{Th. 13.3.1.5(1)} \\
2 & 5 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(x)) = P_{\mathbb{Z}}(S_{\mathbb{Z}}(y)) = E(2) \\
1,2 & 6 \ x = y \quad = E(3, 4, 5)
\end{array}$$

Surjektivität

Th. 13.3.1.7(ii)

$$y \in \mathbb{Z} \vdash \exists x \in \mathbb{Z} S_{\mathbb{Z}}(x) = y$$

$$\begin{array}{ll}
1 & 1 \ y \in \mathbb{Z} \quad A \\
1 & 2 \ P_{\mathbb{Z}}(y) \in \mathbb{Z} \quad \text{Th. 5.2.3.10(Th. 13.3.1.2,1)} \\
1 & 3 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(y)) = y \quad \text{Th. 13.3.1.6(1)} \\
1 & 4 \ \exists x \in \mathbb{Z} S_{\mathbb{Z}}(x) = y \exists I(\wedge I(2, 3))
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 & S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \quad \text{Th. 13.3.1.1} \\
2 & \forall x, y \in \mathbb{Z} (S_{\mathbb{Z}}(x) = S_{\mathbb{Z}}(y) \rightarrow x = y) \text{Th. 13.3.1.7(i)} \\
3 & S_{\mathbb{Z}}: \mathbb{Z} \hookrightarrow \mathbb{Z} \quad \text{Def. 6.2.1.1(1,2)} \\
4 & \forall y \in \mathbb{Z} \exists x \in \mathbb{Z} S_{\mathbb{Z}}(x) = y \quad \text{Th. 13.3.1.7(ii)} \\
5 & S_{\mathbb{Z}}: \mathbb{Z} \twoheadrightarrow \mathbb{Z} \quad \text{Def. 7.2.1.1(1,4)} \\
6 & S_{\mathbb{Z}}: \mathbb{Z} \xrightarrow{\sim} \mathbb{Z} \quad \text{Th. 8.2.1.3(3,5)}
\end{array}$$

□

3.2 Normalformen

Theorem 13.3.2.1 (Positiver Strahl und Nachfolger).

Mengen: n

$$n \in \mathbb{N} \vdash S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)) = \iota_{\mathbb{Z}}(\text{succ}(n))$$

$$\begin{array}{ll}
1 & 1 \ n \in \mathbb{N} \quad A \\
1 & 2 \ S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)) = \iota_{\mathbb{Z}}(n) + 1_{\mathbb{Z}} \quad \text{Def. 13.3.1.1(Th. 5.2.3.10(Th. 13.2.3.1, 1))} \\
3 & 1_{\mathbb{Z}} = \iota_{\mathbb{Z}}(1) \quad \text{Def. 13.2.3.3} \\
1 & 4 \ \iota_{\mathbb{Z}}(n) + \iota_{\mathbb{Z}}(1) = \iota_{\mathbb{Z}}(n + 1) \quad \text{Th. 2.9.1.1(Th. 13.2.6.1(1, Th. 12.4.4.11))} \\
1 & 5 \ n + 1 = \text{succ}(n) \quad \text{Th. 12.4.4.12(1)} \\
1 & 6 \ S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)) = \iota_{\mathbb{Z}}(\text{succ}(n)) \quad = E(2, 3, 4, 5)
\end{array}$$

Theorem 13.3.2.2 (Negativer Strahl und Vorgänger).

Mengen: n

$$n \in \mathbb{N} \vdash P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n)) = -\iota_{\mathbb{Z}}(\text{succ}(n))$$

1 1	$n \in \mathbb{N}$	A
1 2	$\iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}$	$Def. 13.2.3.1(1)$
1 3	$-\iota_{\mathbb{Z}}(n) = [0, n]_{\mathbb{Z}}$	$Def. 13.2.4.1(1, 2)$
1 4	$P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n)) = [0, \text{succ}(n)]_{\mathbb{Z}}$	$Th. 13.3.1.4(1, 3)$
1 5	$\iota_{\mathbb{Z}}(\text{succ}(n)) = [\text{succ}(n), 0]_{\mathbb{Z}}$	$Def. 13.2.3.1(Ax. 12.3.4(1))$
1 6	$-\iota_{\mathbb{Z}}(\text{succ}(n)) = [0, \text{succ}(n)]_{\mathbb{Z}}$	$Def. 13.2.4.1(1, 5)$
1 7	$P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n)) = -\iota_{\mathbb{Z}}(\text{succ}(n))$	$Th. 2.9.1.3(4, 6)$

Theorem 13.3.2.3 (Normalform ganzer Zahlen).

Mengen: z, a, b, n

$$z \in \mathbb{Z} \vdash \exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n))$$

Beweis.

Fall $b \leq a$

Th. 13.3.2.3(i)

$$\begin{aligned} a, b \in \mathbb{N}, b \leq a \vdash \\ \exists n \in \mathbb{N} [a, b]_{\mathbb{Z}} = \iota_{\mathbb{Z}}(n) \end{aligned}$$

1 1	$a, b \in \mathbb{N}$	A
2 2	$b \leq a$	A
1,2 3	$\exists n \in \mathbb{N} (b + n = a)$	$Th. 12.4.5.6(1,2)$
4 4	$n \in \mathbb{N} \wedge b + n = a$	A
1,4 5	$a + 0 = b + n$	$= E(Th. 12.4.4.2(1), \wedge E2(4))$
1,4 6	$[a, b]_{\mathbb{Z}} = [n, 0]_{\mathbb{Z}}$	$Th. 13.2.2.2(1,4,5)$
4 7	$\iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}$	$Def. 13.2.3.1(4)$
1,4 8	$[a, b]_{\mathbb{Z}} = \iota_{\mathbb{Z}}(n)$	$Th. 2.9.1.3(6,7)$
1,2 9	$\exists n \in \mathbb{N} [a, b]_{\mathbb{Z}} = \iota_{\mathbb{Z}}(n)$	$\exists E(3, 4, 8)$

Fall $a \leq b$

Th. 13.3.2.3(ii)

$$\begin{aligned} a, b \in \mathbb{N}, a \leq b \vdash \\ \exists n \in \mathbb{N} [a, b]_{\mathbb{Z}} = -\iota_{\mathbb{Z}}(n) \end{aligned}$$

1	1	$a, b \in \mathbb{N}$	A
2	2	$a \leq b$	A
1,2	3	$\exists n \in \mathbb{N} (a + n = b)$	Th. 12.4.5.6(1,2)
4	4	$n \in \mathbb{N} \wedge a + n = b$	A
1,4	5	$a + n = b + 0$	$= E(\wedge E2(4), Th. 12.4.4.2(1))$
1,4	6	$[a, b]_{\mathbb{Z}} = [0, n]_{\mathbb{Z}}$	Th. 13.2.2.2(1,4,5)
4	7	$\iota_{\mathbb{Z}}(n) = [n, 0]_{\mathbb{Z}}$	Def. 13.2.3.1(4)
4	8	$-\iota_{\mathbb{Z}}(n) = [0, n]_{\mathbb{Z}}$	Def. 13.2.4.1(4,7)
1,4	9	$[a, b]_{\mathbb{Z}} = -\iota_{\mathbb{Z}}(n)$	Th. 2.9.1.3(6,8)
1,2	10	$\exists n \in \mathbb{N} [a, b]_{\mathbb{Z}} = -\iota_{\mathbb{Z}}(n) \exists E(3, 4, 9)$	

Schluss:

1	1	$z \in \mathbb{Z}$	A
1	2	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(1)
3	3	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	4	$b \leq a \vee a \leq b$	Th. 12.4.5.22(3)
5	5	$b \leq a$	A
3,5	6	$\exists n \in \mathbb{N} [a, b]_{\mathbb{Z}} = \iota_{\mathbb{Z}}(n)$	Th. 13.3.2.3(i)(3,5)
3,5	7	$\exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) = E(3, 6)$	
8	8	$a \leq b$	A
3,8	9	$\exists n \in \mathbb{N} [a, b]_{\mathbb{Z}} = -\iota_{\mathbb{Z}}(n)$	Th. 13.3.2.3(ii)(3,8)
3,8	10	$\exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) = E(3, 9)$	
3	11	$\exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) \vee E(4, 5, 7, 8, 10)$	
1	12	$\exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) \exists E(2, 3, 11)$	

□

Kapitel 4

Ordnung der ganzen Zahlen

4.1 Translationsinvarianz der natürlichen Ordnung

Theorem 13.4.1.1 (Translationsinvarianz der natürlichen Ordnung).

Mengen: a, b, c, k

$$a, b, c \in \mathbb{N} \vdash a \leq b \leftrightarrow a + c \leq b + c$$

Beweis.

Erhaltung unter Translation

Th. 13.4.1.1(i)

$$a, b, c \in \mathbb{N}, a \leq b \vdash a + c \leq b + c$$

1	$1 \ a, b, c \in \mathbb{N}$	A
2	$2 \ a \leq b$	A
1,2	$3 \ \exists k \in \mathbb{N} (a + k = b)$	Th. 12.4.5.6(1,2)
4	$4 \ k \in \mathbb{N} \wedge a + k = b$	A
1,4	$5 \ (a + c) + k = (a + k) + c$	Th. 12.4.4.10(1)
1,4	$6 \ (a + k) + c = b + c$	$= E(\wedge E2(4))$
1,4	$7 \ (a + c) + k = b + c$	Th. 2.9.1.2(5,6)
1,4	$8 \ a + c \leq b + c$	Th. 12.4.5.6(1, $\exists I(\wedge I(\wedge E1(4), 7))$)
1,2	$9 \ a + c \leq b + c$	$\exists E(3, 4, 8)$

Kürzen einer Translation

Th. 13.4.1.1(ii)

$$a, b, c \in \mathbb{N}, a + c \leq b + c \vdash a \leq b$$

1	$1 \ a, b, c \in \mathbb{N}$	A
---	------------------------------	-----

2	$2 a + c \leq b + c$	A
1,2	$3 \exists k \in \mathbb{N} ((a + c) + k = b + c)$	Th. 12.4.5.6(1,2)
4	$4 k \in \mathbb{N} \wedge (a + c) + k = b + c$	A
1,4	$5 (a + k) + c = (a + c) + k$	Th. 12.4.4.10(1)
1,4	$6 (a + c) + k = b + c$	$\wedge E2(4)$
1,4	$7 (a + k) + c = b + c$	Th. 2.9.1.2(5,6)
1,4	$8 a + k = b$	Th. 12.4.5.30(Th. 12.4.4.1(1,4),1,1,7)
1,4	$9 a \leq b$	Th. 12.4.5.6(1, $\exists I(\wedge I(\wedge E1(4), 8))$)
1,2	$10 a \leq b$	$\exists E(3, 4, 9)$

Schluss:

1	$1 a, b, c \in \mathbb{N}$	A
1	$2 a \leq b \rightarrow a + c \leq b + c$	Th. 13.4.1.1(i)(1)
1	$3 a + c \leq b + c \rightarrow a \leq b$	Th. 13.4.1.1(ii)(1)
1	$4 a \leq b \leftrightarrow a + c \leq b + c \leftrightarrow I(2, 3)$	

□

Theorem 13.4.1.2 (Addition natürlicher Ungleichungen).

Mengen: a, b, c, d

$$a, b, c, d \in \mathbb{N}, a \leq b, c \leq d \vdash a + c \leq b + d$$

Beweis.

Translation der ersten Ungleichung

Th. 13.4.1.2(i)

$$a, b, c \in \mathbb{N}, a \leq b \vdash a + c \leq b + c$$

1	$1 a, b, c \in \mathbb{N}$	A
2	$2 a \leq b$	A
1,2	$3 a + c \leq b + c$	Th. 13.4.1.1(i)(1,2)

Translation der zweiten Ungleichung

Th. 13.4.1.2(ii)

$$b, c, d \in \mathbb{N}, c \leq d \vdash b + c \leq b + d$$

1	$1 b, c, d \in \mathbb{N}$	A
2	$2 c \leq d$	A
1,2	$3 c + b \leq d + b$	Th. 13.4.1.1(i)(1,2)
1	$4 b + c = c + b$	Th. 12.4.4.8(1)

$$\begin{array}{l} 1 \quad 5 \quad d + b = b + d \text{ Th. 12.4.4.8(1)} \\ 1,2 \quad 6 \quad b + c \leq b + d = E(4, 3, 5) \end{array}$$

Verknüpfung durch Transitivität

Th. 13.4.1.2(iii)

$$a, b, c, d \in \mathbb{N}, \quad a \leq b, \quad c \leq d \vdash a + c \leq b + d$$

$$\begin{array}{l} 1 \quad 1 \quad a, b, c, d \in \mathbb{N} \quad A \\ 2 \quad 2 \quad a \leq b \quad A \\ 3 \quad 3 \quad c \leq d \quad A \\ 1,2 \quad 4 \quad a + c \leq b + c \text{ Th. 13.4.1.2(i)(1,2)} \\ 1,3 \quad 5 \quad b + c \leq b + d \text{ Th. 13.4.1.2(ii)(1,3)} \\ 1 \quad 6 \quad a + c \in \mathbb{N} \quad \text{Th. 12.4.4.1(1)} \\ 1 \quad 7 \quad b + c \in \mathbb{N} \quad \text{Th. 12.4.4.1(1)} \\ 1 \quad 8 \quad b + d \in \mathbb{N} \quad \text{Th. 12.4.4.1(1)} \\ 1,2,3 \quad 9 \quad a + c \leq b + d \text{ Th. 12.4.5.24(6,7,8,4,5)} \end{array}$$

□

4.2 Quotientenordnung

Theorem 13.4.2.1 (Repräsentantenunabhängigkeit der Ganzzahlordnung).

Mengen: $a, b, a', b', c, d, c', d'$

$$\begin{array}{l} a, b, a', b', c, d, c', d' \in \mathbb{N}, \quad [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}}, \quad [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \vdash \\ (a + d \leq c + b) \leftrightarrow (a' + d' \leq c' + b') \end{array}$$

Beweis.

Hinrichtung

Th. 13.4.2.1(i)

$$\begin{array}{l} a, b, a', b', c, d, c', d' \in \mathbb{N}, \quad a + b' = b + a', \quad c + d' = d + c', \quad a + d \leq c + b \vdash \\ a' + d' \leq c' + b' \end{array}$$

$$\begin{array}{ll} 1 \quad 1 \quad a, b, a', b', c, d, c', d' \in \mathbb{N} & A \\ 2 \quad 2 \quad a + b' = b + a' & A \\ 3 \quad 3 \quad c + d' = d + c' & A \\ 4 \quad 4 \quad a + d \leq c + b & A \end{array}$$

$$\begin{array}{ll}
1,4 & 5 \ (a+d) + (b'+c') \leq (c+b) + (b'+c') \text{ Th. 13.4.1.1(i)(1,4)} \\
1,2,3 & 6 \ (a+d) + (b'+c') = (a'+d') + (b+c) \text{ Th. 12.4.4.10(1,2,3)} \\
1 & 7 \ (c+b) + (b'+c') = (c'+b') + (b+c) \text{ Th. 12.4.4.10(1)} \\
1,2,3,4 & 8 \ (a'+d') + (b+c) \leq (c'+b') + (b+c) = E(5,6,7) \\
1,2,3,4 & 9 \ a' + d' \leq c' + b' \text{ Th. 13.4.1.1(ii)(1,8)}
\end{array}$$

Rückrichtung

Th. 13.4.2.1(ii)

$$\begin{array}{l}
a, b, a', b', c, d, c', d' \in \mathbb{N}, \ a + b' = b + a', \ c + d' = d + c', \ a' + d' \leq c' + b' \vdash \\
a + d \leq c + b
\end{array}$$

$$\begin{array}{ll}
1 & 1 \ a, b, a', b', c, d, c', d' \in \mathbb{N} \quad A \\
2 & 2 \ a + b' = b + a' \quad A \\
3 & 3 \ c + d' = d + c' \quad A \\
4 & 4 \ a' + d' \leq c' + b' \quad A \\
1,2 & 5 \ a' + b = b' + a \quad \text{Th. 2.9.1.1(Th. 12.4.4.8(2))} \\
1,3 & 6 \ c' + d = d' + c \quad \text{Th. 2.9.1.1(Th. 12.4.4.8(3))} \\
1,2,3,4 & 7 \ a + d \leq c + b \quad \text{Th. 13.4.2.1(i)(1,5,6,4)}
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 & 1 \ a, b, a', b', c, d, c', d' \in \mathbb{N} \quad A \\
2 & 2 \ [a, b]_{\mathbb{Z}} = [a', b']_{\mathbb{Z}} \quad A \\
3 & 3 \ [c, d]_{\mathbb{Z}} = [c', d']_{\mathbb{Z}} \quad A \\
1,2 & 4 \ a + b' = b + a' \quad \text{Th. 13.2.2.2(1,2)} \\
1,3 & 5 \ c + d' = d + c' \quad \text{Th. 13.2.2.2(1,3)} \\
1,2,3 & 6 \ a + d \leq c + b \rightarrow a' + d' \leq c' + b' \quad \text{Th. 13.4.2.1(i)(1,4,5)} \\
1,2,3 & 7 \ a' + d' \leq c' + b' \rightarrow a + d \leq c + b \quad \text{Th. 13.4.2.1(ii)(1,4,5)} \\
1,2,3 & 8 \ (a + d \leq c + b) \leftrightarrow (a' + d' \leq c' + b') \leftrightarrow I(6,7)
\end{array}$$

□

Definition 13.4.2.1 (Ordnungsoperator der ganzen Zahlen).

Mengen: a, b, c, d
Relationsoperatoren: $\leq_{\mathbb{Z}}$
Neue Symbole: $\leq_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} \leq_{\mathbb{Z}} [c, d]_{\mathbb{Z}} := a + d \leq c + b$$

Definition 13.4.2.2 (Notation der Ganzzahlordnung).

Mengen: z, w
Relationsoperatoren: $\leq, \leq_{\mathbb{Z}}$

$$z, w \in \mathbb{Z} \vdash z \leq w := z \leq_{\mathbb{Z}} w$$

Theorem 13.4.2.2 (Klassencharakterisierung der Ganzzahlordnung).

Mengen: a, b, c, d

Relationsoperatoren: $\leq, \leq_{\mathbb{Z}}$

$$a, b, c, d \in \mathbb{N} \vdash [a, b]_{\mathbb{Z}} \leq [c, d]_{\mathbb{Z}} \leftrightarrow a + d \leq c + b$$

1	1	$a, b, c, d \in \mathbb{N}$	A
1	2	$[a, b]_{\mathbb{Z}} \in \mathbb{Z}$	<i>Th.</i> 13.2.2.1(1)
1	3	$[c, d]_{\mathbb{Z}} \in \mathbb{Z}$	<i>Th.</i> 13.2.2.1(1)
1	4	$[a, b]_{\mathbb{Z}} \leq [c, d]_{\mathbb{Z}} \leftrightarrow [a, b]_{\mathbb{Z}} \leq_{\mathbb{Z}} [c, d]_{\mathbb{Z}}$	<i>Def.</i> 13.4.2.2(2, 3)
1	5	$[a, b]_{\mathbb{Z}} \leq_{\mathbb{Z}} [c, d]_{\mathbb{Z}} \leftrightarrow a + d \leq c + b$	<i>Def.</i> 13.4.2.1(1)
1	6	$[a, b]_{\mathbb{Z}} \leq [c, d]_{\mathbb{Z}} \leftrightarrow a + d \leq c + b$	<i>Th.</i> 2.2.3.5(4, 5)

Theorem 13.4.2.3 (Totale Ordnung der ganzen Zahlen).

Mengen: z, w, u

Relationsoperatoren: $\leq_{\mathbb{Z}}$

$$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$$

Beweis.

Reflexivität

Th. 13.4.2.3(i)

$$z \in \mathbb{Z} \vdash z \leq z$$

1	1	$z \in \mathbb{Z}$	A
1	2	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	<i>Th.</i> 13.2.2.3(1)
3	3	$a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	4	$a + b \leq a + b$	<i>Th.</i> 12.4.5.10(<i>Th.</i> 12.4.4.1(3))
3	5	$[a, b]_{\mathbb{Z}} \leq [a, b]_{\mathbb{Z}}$	<i>Th.</i> 13.4.2.2(3,4)
3	6	$z \leq z$	$= E(3, 5)$
1	7	$z \leq z$	$\exists E(2, 3, 6)$

Antisymmetrie

Th. 13.4.2.3(ii)

$$z, w \in \mathbb{Z}, z \leq w, w \leq z \vdash z = w$$

1	1	$z, w \in \mathbb{Z}$	A
2	2	$z \leq w$	A
3	3	$w \leq z$	A
1	4	$z \in \mathbb{Z}$	$\wedge E1(1)$

1	5	$w \in \mathbb{Z}$	$\wedge E2(1)$
1	6	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(4)
1	7	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.3(5)
8	8	$a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A$	
2,8	9	$a + d \leq c + b$	Th. 13.4.2.2(8,2)
3,8	10	$c + b \leq a + d$	Th. 13.4.2.2(8,3)
8	11	$a + d \in \mathbb{N}$	Th. 12.4.4.1(8)
8	12	$c + b \in \mathbb{N}$	Th. 12.4.4.1(8)
2,3,8	13	$a + d = c + b$	Th. 12.4.5.25(11,12,9,10)
2,3,8	14	$[a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.2(8,13)
2,3,8	15	$z = w$	$= E(8, 14)$
1,2,3	16	$z = w$	$\exists E(6, 7, 8, 15)$

Transitivität

Th. 13.4.2.3(iii)

$$z, w, u \in \mathbb{Z}, z \leq w, w \leq u \vdash z \leq u$$

1	1	$z, w, u \in \mathbb{Z}$	A
2	2	$z \leq w$	A
3	3	$w \leq u$	A
1	4	$z \in \mathbb{Z}$	$\wedge E1(1)$
1	5	$w \in \mathbb{Z}$	$\wedge E1(\wedge E2(1))$
1	6	$u \in \mathbb{Z}$	$\wedge E2(\wedge E2(1))$
4	7	$\exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(4)
5	8	$\exists c, d \in \mathbb{N} w = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.3(5)
6	9	$\exists e, f \in \mathbb{N} u = [e, f]_{\mathbb{Z}}$	Th. 13.2.2.3(6)
10	10	$a, b, c, d, e, f \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} \wedge u = [e, f]_{\mathbb{Z}} A$	
2,10	11	$a + d \leq c + b$	Th. 13.4.2.2(10,2)
3,10	12	$c + f \leq e + d$	Th. 13.4.2.2(10,3)
2,3,10	13	$(a + d) + (c + f) \leq (c + b) + (e + d)$	Th. 13.4.1.2(10,11,12)
10	14	$(a + d) + (c + f) = (a + f) + (c + d)$	Th. 12.4.4.10(10)
10	15	$(c + b) + (e + d) = (b + e) + (c + d)$	Th. 12.4.4.10(10)
2,3,10	16	$(a + f) + (c + d) \leq (b + e) + (c + d)$	$= E(13, 14, 15)$
2,3,10	17	$a + f \leq b + e$	Th. 13.4.1.1(ii)(10,16)
2,3,10	18	$z \leq u$	Th. 13.4.2.2(10,17)
1,2,3	19	$z \leq u$	$\exists E(7, 8, 9, 10, 18)$

Totalität

Th. 13.4.2.3(iv)

$$z, w \in \mathbb{Z} \vdash z \leq w \vee w \leq z$$

1	1	$z, w \in \mathbb{Z}$	A
---	---	-----------------------	-----

1	$2 \ z \in \mathbb{Z}$	$\wedge E1(1)$
1	$3 \ w \in \mathbb{Z}$	$\wedge E2(1)$
2	$4 \ \exists a, b \in \mathbb{N} \ z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(2)
3	$5 \ \exists c, d \in \mathbb{N} \ w = [c, d]_{\mathbb{Z}}$	Th. 13.2.2.3(3)
6	$6 \ a, b, c, d \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}} \wedge w = [c, d]_{\mathbb{Z}} A$	
6	$7 \ a + d \leq c + b \vee c + b \leq a + d$	Th. 12.4.5.22(Th. 12.4.4.1(6), Th. 12.4.4.1(6))
6	$8 \ z \leq w \vee w \leq z$	Th. 13.4.2.2(6,7)
1	$9 \ z \leq w \vee w \leq z$	$\exists E(4, 5, 6, 8)$

Schluss:

1	$\forall z \in \mathbb{Z} (z \leq z)$	Th. 13.4.2.3(i)
2	$2 \ z \in \mathbb{Z}$	A
3	$3 \ w \in \mathbb{Z}$	A
4	$4 \ u \in \mathbb{Z}$	A
5	$5 \ z \leq w$	A
6	$6 \ w \leq u$	A
2,3,4,5,6	$7 \ z \leq u$	Th. 13.4.2.3(iii)(2,3,4,5,6)
8	$8 \ w \leq z$	A
2,3,5,8	$9 \ z = w$	Th. 13.4.2.3(ii)(2,3,5,8)
10	$\text{PartOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$	Def. 11.2.1.1(1,7,9)
2,3	$11 \ z \leq w \vee w \leq z$	Th. 13.4.2.3(iv)(2,3)
12	$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$	Def. 11.2.1.3(10,11)

□

Definition 13.4.2.3 (Strikte Ordnung der ganzen Zahlen).

Mengen: z, w

Relationsoperatoren: $<$

$$z, w \in \mathbb{Z} \vdash z < w := z \leq w \wedge z \neq w$$

Theorem 13.4.2.4 (Jede ganze Zahl ist kleiner als ihr Nachfolger).

Mengen: z, a, b

$$z \in \mathbb{Z} \vdash z < S_{\mathbb{Z}}(z)$$

Beweis.

Ordnungsteil

Th. 13.4.2.4(i)

$$z \in \mathbb{Z} \vdash z \leq S_{\mathbb{Z}}(z)$$

1	$z \in \mathbb{Z}$	A
1	$2 \exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(1)
3	$3 a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
3	$4 S_{\mathbb{Z}}(z) = [\text{succ}(a), b]_{\mathbb{Z}}$	Th. 13.3.1.3(3)
3	$5 a \leq \text{succ}(a)$	Th. 12.4.5.12(3)
3	$6 a + b \leq \text{succ}(a) + b$	Th. 13.4.1.1(i)(3,5)
3	$7 z \leq S_{\mathbb{Z}}(z)$	Th. 13.4.2.2(3,4,6)
1	$8 z \leq S_{\mathbb{Z}}(z)$	$\exists E(2, 3, 7)$

Ungleichheitsteil

Th. 13.4.2.4(ii)

$$z \in \mathbb{Z} \vdash z \neq S_{\mathbb{Z}}(z)$$

1	$1 z \in \mathbb{Z}$	A
2	$2 z = S_{\mathbb{Z}}(z)$	A
1	$3 \exists a, b \in \mathbb{N} z = [a, b]_{\mathbb{Z}}$	Th. 13.2.2.3(1)
4	$4 a, b \in \mathbb{N} \wedge z = [a, b]_{\mathbb{Z}}$	A
4	$5 S_{\mathbb{Z}}(z) = [\text{succ}(a), b]_{\mathbb{Z}}$	Th. 13.3.1.3(4)
2,4	$6 [a, b]_{\mathbb{Z}} = [\text{succ}(a), b]_{\mathbb{Z}}$	$= E(4, 5, 2)$
2,4	$7 a + b = b + \text{succ}(a)$	Th. 13.2.2.2(4,6)
4	$8 b + \text{succ}(a) = \text{succ}(a) + b$	Th. 12.4.4.8(4)
2,4	$9 a + b = \text{succ}(a) + b$	$= E(7, 8)$
2,4	$10 a = \text{succ}(a)$	Th. 12.4.5.30(4,9)
4	$11 a < \text{succ}(a)$	Th. 12.4.5.13(4)
2,4	$12 \perp$	Th. 12.4.5.28(4,11,10)
1,2	$13 \perp$	$\exists E(3, 4, 12)$
1	$14 z \neq S_{\mathbb{Z}}(z)$	$\neg I(2, 13)$

Schluss:

1	$1 z \in \mathbb{Z}$	A
1	$2 z \leq S_{\mathbb{Z}}(z)$	Th. 13.4.2.4(i)(1)
1	$3 z \neq S_{\mathbb{Z}}(z)$	Th. 13.4.2.4(ii)(1)
1	$4 z < S_{\mathbb{Z}}(z)$	Def. 13.4.2.3(1, $\wedge I(2, 3)$)

□

Kapitel 5

Erweiterte Peano-Axiome der ganzen Zahlen

5.1 Herleitung der zweiseitigen Induktion

Theorem 13.5.1.1 (Zweiseitige Induktion im Quotientenmodell).

Einstelliges Prädikat: P
Mengen: z, n

$$P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))), \\ \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \forall z \in \mathbb{Z} P(z)$$

Beweis.

Induktion auf dem nichtnegativen Strahl

Th. 13.5.1.1(i)

$$P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))) \vdash \\ \forall n \in \mathbb{N} P(\iota_{\mathbb{Z}}(n))$$

1	1	$P(0_{\mathbb{Z}})$	A
2	2	$\forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z)))$	A
	3	$0_{\mathbb{Z}} = \iota_{\mathbb{Z}}(0)$	Def. 13.2.3.2
1	4	$P(\iota_{\mathbb{Z}}(0))$	$= E(3, 1)$
5	5	$n \in \mathbb{N}$	A
6	6	$P(\iota_{\mathbb{Z}}(n))$	A
5	7	$\iota_{\mathbb{Z}}(n) \in \mathbb{Z}$	Th. 5.2.3.10(Th. 13.2.3.1,5)
2,5	8	$P(\iota_{\mathbb{Z}}(n)) \rightarrow P(S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)))$	$\rightarrow E(\forall E(2), 7)$
2,5,6	9	$P(S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)))$	$\rightarrow E(8, 6)$
5	10	$S_{\mathbb{Z}}(\iota_{\mathbb{Z}}(n)) = \iota_{\mathbb{Z}}(\text{succ}(n))$	Th. 13.3.2.1(5)
2,5,6	11	$P(\iota_{\mathbb{Z}}(\text{succ}(n)))$	$= E(10, 9)$

$$\begin{array}{ll}
2,5 \quad 12 \quad P(\iota_{\mathbb{Z}}(n)) \rightarrow P(\iota_{\mathbb{Z}}(\text{succ}(n))) & \rightarrow I(6, 11) \\
2 \quad 13 \quad \forall n \in \mathbb{N} (P(\iota_{\mathbb{Z}}(n)) \rightarrow P(\iota_{\mathbb{Z}}(\text{succ}(n)))) \forall I(\rightarrow I(5, 12)) & \\
1,2 \quad 14 \quad \forall n \in \mathbb{N} P(\iota_{\mathbb{Z}}(n)) & \text{Ax. 12.3.8(4,13)}
\end{array}$$

Induktion auf dem nichtpositiven Strahl

Th. 13.5.1.1(ii)

$$\begin{array}{l}
P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \\
\forall n \in \mathbb{N} P(-\iota_{\mathbb{Z}}(n))
\end{array}$$

$$\begin{array}{ll}
1 \quad 1 \quad P(0_{\mathbb{Z}}) & A \\
2 \quad 2 \quad \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) & A \\
3 \quad \iota_{\mathbb{Z}}(0) = 0_{\mathbb{Z}} & \text{Th. 2.9.1.1(Def. 13.2.3.2)} \\
4 \quad -0_{\mathbb{Z}} = 0_{\mathbb{Z}} & \text{Th. 13.2.4.12} \\
1 \quad 5 \quad P(-\iota_{\mathbb{Z}}(0)) & = E(3, 4, 1) \\
6 \quad 6 \quad n \in \mathbb{N} & A \\
7 \quad 7 \quad P(-\iota_{\mathbb{Z}}(n)) & A \\
6 \quad 8 \quad -\iota_{\mathbb{Z}}(n) \in \mathbb{Z} & \text{Th. 13.2.4.3(Th. 5.2.3.10(Th. 13.2.3.1,6))} \\
2,6 \quad 9 \quad P(-\iota_{\mathbb{Z}}(n)) \rightarrow P(P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n))) & \rightarrow E(\forall E(2), 8) \\
2,6,7 \quad 10 \quad P(P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n))) & \rightarrow E(9, 7) \\
6 \quad 11 \quad P_{\mathbb{Z}}(-\iota_{\mathbb{Z}}(n)) = -\iota_{\mathbb{Z}}(\text{succ}(n)) & \text{Th. 13.3.2.2(6)} \\
2,6,7 \quad 12 \quad P(-\iota_{\mathbb{Z}}(\text{succ}(n))) & = E(11, 10) \\
2,6 \quad 13 \quad P(-\iota_{\mathbb{Z}}(n)) \rightarrow P(-\iota_{\mathbb{Z}}(\text{succ}(n))) & \rightarrow I(7, 12) \\
2 \quad 14 \quad \forall n \in \mathbb{N} (P(-\iota_{\mathbb{Z}}(n)) \rightarrow P(-\iota_{\mathbb{Z}}(\text{succ}(n)))) \forall I(\rightarrow I(6, 13)) & \\
1,2 \quad 15 \quad \forall n \in \mathbb{N} P(-\iota_{\mathbb{Z}}(n)) & \text{Ax. 12.3.8(5,14)}
\end{array}$$

Schluss mittels Normalform:

$$\begin{array}{ll}
1 \quad 1 \quad P(0_{\mathbb{Z}}) & A \\
2 \quad 2 \quad \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))) & A \\
3 \quad 3 \quad \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) & A \\
1,2 \quad 4 \quad \forall n \in \mathbb{N} P(\iota_{\mathbb{Z}}(n)) & \text{Th. 13.5.1.1(i)(1,2)} \\
1,3 \quad 5 \quad \forall n \in \mathbb{N} P(-\iota_{\mathbb{Z}}(n)) & \text{Th. 13.5.1.1(ii)(1,3)} \\
6 \quad 6 \quad z \in \mathbb{Z} & A \\
6 \quad 7 \quad \exists n \in \mathbb{N} (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) & \text{Th. 13.3.2.3(6)} \\
8 \quad 8 \quad n \in \mathbb{N} \wedge (z = \iota_{\mathbb{Z}}(n) \vee z = -\iota_{\mathbb{Z}}(n)) & A \\
4,8 \quad 9 \quad P(\iota_{\mathbb{Z}}(n)) & \rightarrow E(\forall E(4), \wedge E1(8)) \\
5,8 \quad 10 \quad P(-\iota_{\mathbb{Z}}(n)) & \rightarrow E(\forall E(5), \wedge E1(8)) \\
11 \quad 11 \quad z = \iota_{\mathbb{Z}}(n) & A \\
4,8,11 \quad 12 \quad P(z) & = E(11, 9) \\
13 \quad 13 \quad z = -\iota_{\mathbb{Z}}(n) & A \\
5,8,13 \quad 14 \quad P(z) & = E(13, 10) \\
4,5,8 \quad 15 \quad P(z) & \vee E(\wedge E2(8), 11, 12, 13, 14) \\
4,5,6 \quad 16 \quad P(z) & \exists E(7, 8, 15)
\end{array}$$

$$1,2,3 \quad 17 \quad \forall z \in \mathbb{Z} P(z)$$

$$\forall I(\rightarrow I(6, 16))$$

□

5.2 Axiomatische Zusammenfassung

Die folgenden Axiome erweitern die Peano-Struktur um einen überall definierten Vorgänger. Anders als auf $\mathbb{N}$ ist der Nachfolger auf $\mathbb{Z}$ bijektiv. Die totale Ordnung und die strikte Nachfolgerorientierung verhindern zyklische Nachfolgermodelle; die zweiseitige Induktion verbindet den positiven und den negativen Strahl.

Metalogisch ist dabei zwischen zwei Lesarten zu unterscheiden: Als erststufiges Axiomenschema ist die Beschreibung nicht kategorisch und besitzt nichtstandardmäßige Modelle. Unter voller zweitstufiger Lesart der Induktion beschreibt sie dagegen die intendierte zweiseitige, von $0_{\mathbb{Z}}$ durch Nachfolger und Vorgänger erzeugte Kette.

Definition 13.5.2.1 (Erweiterte Peano-Axiome der ganzen Zahlen).

Axiome:

<i>Null:</i>	$0_{\mathbb{Z}} \in \mathbb{Z}$	<i>Ax. 13.5.2.1</i>
<i>Nachfolger:</i>	$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	<i>Ax. 13.5.2.2</i>
<i>Vorgänger:</i>	$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	<i>Ax. 13.5.2.3</i>
<i>Vorgänger nach Nachfolger:</i>	$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$	<i>Ax. 13.5.2.4</i>
<i>Nachfolger nach Vorgänger:</i>	$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$	<i>Ax. 13.5.2.5</i>
<i>Totale Ordnung:</i>	$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$	<i>Ax. 13.5.2.6</i>
<i>Nachfolgerorientierung:</i>	$z \in \mathbb{Z} \vdash z < S_{\mathbb{Z}}(z)$	<i>Ax. 13.5.2.7</i>
<i>Zweiseitige Induktion:</i>	$P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))),$ $\forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \forall z \in \mathbb{Z} P(z)$	<i>Ax. 13.5.2.8</i>

Neue Symbole:

<i>Ganze Zahlen:</i>	$\mathbb{Z}$
<i>Null:</i>	$0_{\mathbb{Z}}$
<i>Nachfolger:</i>	$S_{\mathbb{Z}}$
<i>Vorgänger:</i>	$P_{\mathbb{Z}}$
<i>Ordnung:</i>	$\leq_{\mathbb{Z}}$

$$\text{ZPeano}(\mathbb{Z}, 0_{\mathbb{Z}}, S_{\mathbb{Z}}, P_{\mathbb{Z}}, \leq_{\mathbb{Z}})$$

Theorem 13.5.2.1 (Das Quotientenmodell erfüllt die Ganzzahlaxiome).

<i>Mengen:</i>	$z, D_{\mathbb{Z}}$
<i>Funktionen:</i>	$S_{\mathbb{Z}}, P_{\mathbb{Z}}$
<i>Relationsoperatoren:</i>	$\sim_{\mathbb{Z}}, \leq_{\mathbb{Z}}$

$$\mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}} \wedge$$

$$\text{ZPeano}(\mathbb{Z}, 0_{\mathbb{Z}}, S_{\mathbb{Z}}, P_{\mathbb{Z}}, \leq_{\mathbb{Z}})$$

Beweis.

Träger und Abbildungen

Th. 13.5.2.1(i)

$$0_{\mathbb{Z}} \in \mathbb{Z} \wedge S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \wedge \\ P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$\begin{array}{ll} 1 \ 0_{\mathbb{Z}} \in \mathbb{Z} & \text{Th. 13.2.3.4} \\ 2 \ S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} & \text{Th. 13.3.1.1} \\ 3 \ P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} & \text{Th. 13.3.1.2} \\ 4 \ 0_{\mathbb{Z}} \in \mathbb{Z} \wedge S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \wedge P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z} \wedge I(1, \wedge I(2, 3)) \end{array}$$

Gegenseitige Inversen

Th. 13.5.2.1(ii)

$$\forall z \in \mathbb{Z} \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z \wedge \\ \forall z \in \mathbb{Z} \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$$

$$\begin{array}{ll} 1 \ \forall z \in \mathbb{Z} \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z & \text{Th. 13.3.1.5} \\ 2 \ \forall z \in \mathbb{Z} \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z & \text{Th. 13.3.1.6} \\ 3 \ \forall z \in \mathbb{Z} \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z \wedge \forall z \in \mathbb{Z} \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z \wedge I(1, 2) \end{array}$$

Geordnete Nachfolgerstruktur

Th. 13.5.2.1(iii)

$$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}}) \wedge \forall z \in \mathbb{Z} \ (z < S_{\mathbb{Z}}(z))$$

$$\begin{array}{ll} 1 \ \text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}}) & \text{Th. 13.4.2.3} \\ 2 \ \forall z \in \mathbb{Z} \ (z < S_{\mathbb{Z}}(z)) & \text{Th. 13.4.2.4} \\ 3 \ \text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}}) \wedge \forall z \in \mathbb{Z} \ (z < S_{\mathbb{Z}}(z)) \wedge I(1, 2) \end{array}$$

Induktionsschema

Th. 13.5.2.1(iv)

$$P(0_{\mathbb{Z}}), \ \forall z \in \mathbb{Z} \ (P(z) \rightarrow P(S_{\mathbb{Z}}(z))), \\ \forall z \in \mathbb{Z} \ (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \forall z \in \mathbb{Z} \ P(z)$$

$$\begin{array}{ll} 1 \ 1 \ P(0_{\mathbb{Z}}) & A \\ 2 \ 2 \ \forall z \in \mathbb{Z} \ (P(z) \rightarrow P(S_{\mathbb{Z}}(z))) A & \\ 3 \ 3 \ \forall z \in \mathbb{Z} \ (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) A & \\ 1,2,3 \ 4 \ \forall z \in \mathbb{Z} \ P(z) & \text{Th. 13.5.1.1(1,2,3)} \end{array}$$

Schluss:

$$1 \ \mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}} \quad \text{Def. 13.2.2.1}$$

2	$0_{\mathbb{Z}} \in \mathbb{Z}$	$\wedge E1(\text{Th. 13.5.2.1}(i))$
3	$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	$\wedge E1(\wedge E2(\text{Th. 13.5.2.1}(i)))$
4	$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$	$\wedge E2(\wedge E2(\text{Th. 13.5.2.1}(i)))$
5	$5 \ z \in \mathbb{Z}$	A
5	$6 \ P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$	$\rightarrow E(5, \wedge E1(\text{Th. 13.5.2.1}(ii)))$
5	$7 \ S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$	$\rightarrow E(5, \wedge E2(\text{Th. 13.5.2.1}(ii)))$
8	$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$	$\wedge E1(\text{Th. 13.5.2.1}(iii))$
5	$9 \ z < S_{\mathbb{Z}}(z)$	$\rightarrow E(5, \wedge E2(\text{Th. 13.5.2.1}(iii)))$
10	$10 \ P(0_{\mathbb{Z}})$	A
11	$11 \ \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z)))$	A
12	$12 \ \forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z)))$	A
10,11,12	$13 \ \forall z \in \mathbb{Z} P(z)$	$\text{Th. 13.5.2.1}(iv)(10,11,12)$
14	$\text{ZPeano}(\mathbb{Z}, 0_{\mathbb{Z}}, S_{\mathbb{Z}}, P_{\mathbb{Z}}, \leq_{\mathbb{Z}})$	$\text{Def. 13.5.2.1}(2,3,4,6,7,8,9,13)$
15	$\mathbb{Z} = D_{\mathbb{Z}} / \sim_{\mathbb{Z}} \wedge \text{ZPeano}(\mathbb{Z}, 0_{\mathbb{Z}}, S_{\mathbb{Z}}, P_{\mathbb{Z}}, \leq_{\mathbb{Z}}) \wedge I(1, 14)$	

□

Axiom 13.5.2.1 (Null-Axiom der ganzen Zahlen).

$$0_{\mathbb{Z}} \in \mathbb{Z}$$

Herkunft aus der Quotientenkonstruktion: Th. 13.2.3.4.

Axiom 13.5.2.2 (Nachfolgerfunktion der ganzen Zahlen).

$$S_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

Herkunft aus der Quotientenkonstruktion: Th. 13.3.1.1.

Axiom 13.5.2.3 (Vorgängerfunktion der ganzen Zahlen).

$$P_{\mathbb{Z}}: \mathbb{Z} \rightarrow \mathbb{Z}$$

Herkunft aus der Quotientenkonstruktion: Th. 13.3.1.2.

Axiom 13.5.2.4 (Vorgänger nach Nachfolger).

$$z \in \mathbb{Z} \vdash P_{\mathbb{Z}}(S_{\mathbb{Z}}(z)) = z$$

Herkunft aus der Quotientenkonstruktion: Th. 13.3.1.5.

Axiom 13.5.2.5 (Nachfolger nach Vorgänger).

$$z \in \mathbb{Z} \vdash S_{\mathbb{Z}}(P_{\mathbb{Z}}(z)) = z$$

Herkunft aus der Quotientenkonstruktion: Th. 13.3.1.6.

Axiom 13.5.2.6 (Totale Ordnung der ganzen Zahlen).

$$\text{TotOrd}(\mathbb{Z}, \leq_{\mathbb{Z}})$$

Herkunft aus der Quotientenkonstruktion: Th. 13.4.2.3.

Axiom 13.5.2.7 (Strikte Nachfolgerorientierung).

$$z \in \mathbb{Z} \vdash z < S_{\mathbb{Z}}(z)$$

Herkunft aus der Quotientenkonstruktion: Th. 13.4.2.4.

Axiom 13.5.2.8 (Zweiseitiges Induktionsaxiom).

$$\begin{aligned} &P(0_{\mathbb{Z}}), \forall z \in \mathbb{Z} (P(z) \rightarrow P(S_{\mathbb{Z}}(z))), \\ &\forall z \in \mathbb{Z} (P(z) \rightarrow P(P_{\mathbb{Z}}(z))) \vdash \forall z \in \mathbb{Z} P(z) \end{aligned}$$

Herkunft aus der Quotientenkonstruktion: Th. 13.5.1.1.